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RESEARCH ARTICLE

Addressing Uncertainty in Online Alarm Flood Classification Using Conformal Prediction

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ABSTRACT Alarm flood management is essential for industrial process plant safety and efficiency. Online “alarm flood classification” (AFC) assigns an observed sequence of alarms to one (of many) alarm flood classes known from the past. Nevertheless, accurately differentiating between alarm flood classes as they evolve over time remains a challenge due to classification uncertainty. Furthermore, AFC usually assigns to just one alarm flood class, preventing the operator from recognizing that multiple classes should be considered given the limited information at hand. This paper proposes a novel approach by integrating “conformal prediction” with “early time series classification” to address these challenges, offering a probabilistic understanding of alarm flood classes to be considered, and enhancing decision-support for operators. Using a method-agnostic framework, our method can be used with existing AFC models, making them adaptable to evolving alarm dynamics. We apply our approach to five AFC methods from literature using a novel and openly accessible synthetic alarm dataset we designed. Our evaluation shows how our method can improve coverage of the existing models, particularly in the early stages of alarm floods, ensuring with high confidence that the true class is captured while preserving a reasonably small set of possible alarm flood classes.

INDEX TERMS Abnormal situations, alarm management, conformal prediction, early time series classification, industrial alarm floods, industrial process diagnosis.

I. INTRODUCTION

In the dynamic environment of industrial process plants, effective alarm management is crucial for ensuring operational safety and efficiency [1], [2], [3]. “Process control systems” (PCSs) are equipped with alarms that are specifically designed to activate when “process variables” (PVs) deviate from predetermined thresholds, such as the pressure within a stripper column. These alarms act as critical indicators, notifying operators of process deviations that require manual intervention [1], [2].

Ideally, the system aims to minimize the activation of alarms to prevent operator overload [1], [2], [3]. Neverthe-

less, in the event of complex abnormal situations, a rapid succession of multiple alarms, known commonly as an alarm flood [2], [3], [4], [5], can occur. Such alarm floods pose significant challenges to operators’ situational awareness, making it difficult to address process deviations promptly and accurately [4]. Notably, alarm floods have been implicated in major industrial incidents [6], [7], [8], underscoring the need for alarm management strategies to mitigate their impact.

Alarm floods, which can occur when disturbances propagate through interconnected sections of a plant, demonstrate the drawbacks of conventional, sequential alarm resolution methods [4], [9]. Critical alarms may not always be activated first due to different alarm thresholds [10]. Consequently, operators are compelled to engage in a time-consuming and

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demanding process of decision-making to assess and resolve the underlying causes of these alarms [1], [2], [3]. Herein lies the potential of “machine learning” (ML) and “artificial intelligence” (AI)-based analyses to extract patterns and knowledge from readily available historical alarm data [11]. These solutions have the potential to offer operators valuable insights and decision-support to accelerate the restoration of normal process operations [5], [12].

Data-driven alarm analysis in industrial settings encompasses different approaches designed to enhance the management and understanding of alarm data [5]. Alarm flood clustering groups similar historical alarm floods together based on their characteristics, enabling engineers to identify common patterns and recurring issues [13]. This clustering provides a basis for labeling historical data, which can be used for training classification models [5], [12]. Alarm similarity analysis examines the correlation between the signals of individual alarms, potentially revealing dependencies between different alarms that could indicate underlying process dynamics [14]. Additionally, alarm flood root-cause analysis focuses on identifying the fundamental causes of alarm floods, providing insights that are relevant for diagnosing and resolving process disturbances [5], [10].

Our study focuses on “alarm flood classification” (AFC) methods, which are an advanced approach to analyzing alarm data [5]. AFC methods categorize alarm floods or, more broadly, alarm subsequences, which are smaller segments of a longer alarm stream [13], [14], [15], [16], by comparing them to historical occurrences and their typical patterns. The latter can be extracted leveraging the labels generated through alarm flood clustering [5], [12]. Consequently, AFC has the ability to predict the most likely alarm subsequence class and provide it to the human operator. This information can be further enhanced utilizing the insights from root-cause analysis and similarity analysis to provide operators with more detailed information, e.g., potential root causes, historical context, and recommended actions [5]. By integrating this information, AFC methods have the potential to reduce the cognitive load on human experts, minimize the risks of manual assessment errors, and enhance decision support for operators in restoring normal operations [5], [17].

Despite recent advancements in AFC, e.g., in [18], [19], and [20], the classification process still faces challenges. As alarm subsequences unfold, the dynamic nature of process plants can lead to temporary class ambiguity. This means that different classes may appear to be indistinguishable for a certain duration, as evidenced by [17] and [19]. The level of ambiguity decreases as more components of the plant are impacted, resulting in the activation of additional alarms and subsequently augmenting the number of distinct dynamics. Consequently, in the initial stages, when only a small number of alarms are triggered, various alarm subsequence classes may closely resemble one another, thereby making the AFC task more error-prone [17], [19].

A similar phenomenon has also been described in the related field of fault detection and diagnosis in [21]. The study

in [21] addresses the issue of classifying gradually developing incipient process faults, which can be difficult at first due to limited information available. However, as these faults propagate and affect more components of the process, they become easier to identify and distinguish.

We contend that the issue of temporary class ambiguity is especially prevalent in smaller plants, where there are only a restricted number of distinct alarms implemented in the PCS that can be used to differentiate between classes, as well as in abnormal situations where similar root-causes result in different outcomes over time. Temporary class ambiguity can be further influenced by alarm subsequences being commonly clustered based on their overall dynamics [5], [13], [22]. Subsequently, these clusters serve as class labels for training AFC models [5], [12]. This approach, however, fails to account for situations where the initial segments of an alarm subsequence do not contain enough characteristic dynamics to clearly distinguish between different classes.

Moreover, beyond temporary class ambiguity, there exist additional aspects of AFC uncertainty stemming from the simplifying assumptions inherent in the models’ design. These simplifications may result in models that do not fully capture the alarm dynamics they intend to classify, leading to an inability to differentiate between classes when utilizing this restricted representation rather than the entire body of available alarm dynamics [12]. Additionally, limitations in the available training data may increase uncertainty due to not presenting enough information for extracting significant patterns. The operator should be informed in situations where a prediction cannot be made with high certainty but requires consideration of multiple potential alarm subsequence classes [23]. These challenges underscore the need for an AFC approach that can manage uncertainty and adapt to the evolving dynamics of alarm floods.

Classification uncertainties and dynamic classification challenges have already been studied across various fields, leading to the development of methods to manage some of these complexities. Uncertainty-aware neural networks generate predictive sets of possible classes, e.g., using Monte Carlo dropout to generate posterior predictive distributions, from which the model uncertainty can be obtained [24]. However, these models require extensive datasets for training, which may not be available in industrial alarm flood scenarios. Moreover, this type of uncertainty-awareness is not method-agnostic and only applicable to neural networks, whereas in AFC various model types are utilized. An alternative approach is to learn multi-label classifications directly through supervised learning [25]. Nevertheless, this necessitates the accessibility of detailed labels. In the case of temporary class ambiguity, multi-class labels would be required for each time step in which we expect the AFC model to make a prediction. We argue that this level of detail might be excessive for practical acquisition.

Another type of methods, ensemble methods, inherently quantify uncertainty by analyzing the variation in predictions among different models [26]. However, they might still be

ineffective in capturing all possible classes, especially if the majority of the models display a preference for a dominant class when confronted with temporary class ambiguity. In addition, time series segmentation-based classification involves dividing the time series into segments, e.g., using sliding windows, and independently classifying each segment [27]. This allows for dynamic classification by giving priority to the investigation of temporary relevant sections. However, if segments are treated independently, there might be a risk of losing the overall context, particularly when dealing with temporary class ambiguity that cannot be resolved by segment level predictions.

“Conformal prediction” (CP) and “early time series classification” (ETSC) are two additional approaches that have been explored to address ambiguity and uncertainty in data and models. CP provides prediction sets that include the true class, by using statistical techniques that calibrate the model’s prediction to maintain the desired coverage probability [23]. This is achieved without requiring additional multi-labels for training and without being dependent on any specific model type, thus overcoming the limitations of traditional multi-label classification and uncertainty-aware neural networks. However, the use of CP in classification is typically limited to static data, e.g., medical images, restricting its effectiveness in dynamic scenarios like AFC, where data continuously evolves. In contrast, ETSC focuses on identifying the optimal point in time for early and accurate classification by employing segmentation techniques that are specifically designed to preserve the overall context [28], thus overcoming the constraints of time series segmentation-based classification techniques such as sliding windows. While ETSC effectively balances timeliness and accuracy, it primarily targets a specific classification point, offering limited support during the periods before or after this point when ambiguity may still be present.

To overcome the limitations of current methods in resolving AFC uncertainty, we suggest integrating CP and ETSC within a method-agnostic framework tailored to online AFC, i.e., the classification of ongoing alarm subsequences [19]. Using CP, we aim to improve existing AFC methods by offering sets of possible subsequence classes when there is significant uncertainty in class distinction. Additionally, by leveraging ETSC, our approach refines the classification process by capturing dynamics unique to specific segments of the alarm subsequence and continuously updating the predicted sets of potential classes, even as new data continuously unfolds. This combination allows for effectively handling the evolving nature of alarm floods and managing ambiguity throughout their entire duration, resulting in a more adaptive and robust AFC that provides operators with timely and reliable decision-support.

Research has demonstrated that the operators’ trust in the alarm system plays a crucial role in their capacity to address abnormal situations effectively [29], [30]. On the other hand, offering premature assistance without sufficient information may undermine their trust [31]. For this reason, the contri-

bution and purpose of our method is twofold. First, we aim to provide human operators with a probabilistic and dynamic understanding of potential alarm subsequence classes to be considered by providing them with an evolving set of potential classes when the model is uncertain, taking into account any ambiguities in the stream of observed alarm data. Second, we aim to achieve a balance between ensuring a high level of confidence in capturing the true alarm subsequence class, thus improving coverage, and maintaining a relatively small set of potential classes. In accordance with [32], we argue that this could enhance the operators’ trust in the alarm system and AI-powered decision support tools. This might also enhance their ability to promptly address emerging abnormal situations by identifying the situations that their actions should prioritize [29].

In a commitment to fostering transparency in industrial AI applications, we have made our code and its usage rights openly accessible. This allows other researchers in the field to test, apply and re-use them in open-source collaboration.

The remainder of this paper is organized as follows: Section II provides a review of the related work. Section III outlines the methodology of our novel approach for enhancing online AFC. In Section IV, we apply our approach to five AFC methods from literature, utilizing a novel and openly accessible synthetic alarm dataset. The results of this evaluation are presented and discussed. Finally, this paper concludes in Section V with the most significant findings and suggestions for future research.

II. RELATED WORK

In this section, we provide a review of the relevant literature. In Subsection II-A, we examine existing AFC methods, describing their overall design, as well as their unique advantages, limitations, and remaining challenges. Next, in Subsection II-B and Subsection II-C, we examine and analyze CP and ETSC, respectively. Subsection II-D concludes by summarizing the main findings of our literature review.

A. ALARM FLOOD CLASSIFICATION

1) REQUIREMENTS AND OVERVIEW

The reviews in [17] and [18] offer thorough analyses of existing methods for AFC. Online AFC, however, lacks sufficient exploration in the literature [19]. A foundational assumption underlying AFC methods is the notion that similar alarm subsequences are indicative of similar abnormal situations. However, there is a lack of agreement on a universally accepted set of requirements for effective AFC, as evidenced by the varying criteria used to define similarity among alarm subsequences [5]. Nonetheless, [17] outlines specific requirements ($R1$ and $R2$) for online AFC:

- R1) AFC methods are applied to alarm subsequences that are evolving over time, with alarms being activated and deactivated. Therefore, online AFC must be conducted incrementally and as early as possible.

R2) Given the potential for different alarm subsequence classes to exhibit temporary resemblance, an effective online AFC method must be capable of providing a multi-label output that includes all plausible classes.

Most AFC methodologies described in the literature follow a multi-class classification approach, using supervised learning techniques that make use of historical alarm data and annotated class labels [18]. This approach generally involves a hierarchical structure consisting of three tiers [5]:

- 1) During the *transformation stage*, key features or characteristics are extracted from the alarm data of an alarm subsequence to be classified.
- 2) The *classification stage* involves matching these extracted features against the most appropriate class of historical alarm subsequences using a classifier.
- 3) A *novelty detection stage* employs a threshold mechanism to ascertain whether the alarm subsequence under classification aligns with a known class or represents a novel, previously unobserved class.

Despite this common approach, AFC methods can be categorized based on their alarm data input [5], i.e., a set, sequence, or series representation. An alarm set is the set of unique alarm variables, which are the unique identifiers for the alarms defined in a PCS [13], that are activated at least once in an alarm subsequence. Alarm sequences are lists of chronologically ordered and time-stamped alarm activations in alarm variables. An alarm series consists of a set of binary time series that show the dynamic activity of individual alarm variables. Alarm sets have the least amount of incorporated alarm dynamics, while alarm series have the most [5], [13]. Fig. 1 shows alarm series (Fig. 1(a)) and sequence (Fig. 1(b)) representations of an alarm subsequence, where the alarm sequence consists of the rising edges of the alarm series' variables. The corresponding alarm set consists of the activated alarm variables A, B, C, and D.

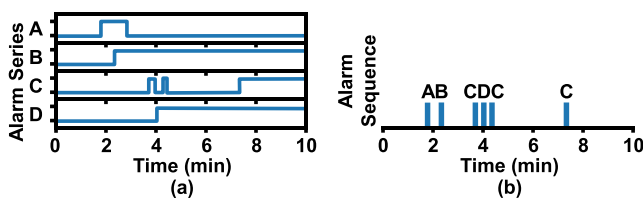


FIGURE 1. Two types of alarm data representations for alarm variables A to D. (a) Alarm series. The solid blue lines represent alarm variable time trends. A higher level represents an active alarm. (b) Alarm sequence. The solid blue lines indicate alarm activations.

2) ALARM SET-BASED METHODS

Alarm set-based AFC approaches commonly employ string metrics to estimate the distance between two alarm subsequences, without focusing on the number or order of alarm activations [5]. In reference [33], an AFC method was introduced that utilizes a weighted dissimilarity index to transform alarm data into binary alarm set vectors. A template alarm set vector is created for each historical alarm subsequence

class. This vector includes only the alarm variables that are active in at least half of the alarm subsequences in the class. The templates generated are further weighted based on the degree of characteristic representation of a certain alarm variable within a class, in comparison to all other classes. For a new alarm subsequence, the differences between it and each of the weighted historical templates are computed. Subsequently, a classification phase is performed using a non-parametric “first-nearest-neighbor” (1NN) classifier. String metrics, however, tend to overstate the similarity between two alarm subsequences that have shared alarms but significant differences in their respective dynamics [13], [33]. This can make them more vulnerable to temporary class ambiguities and lead to high AFC uncertainties (R2). In addition, [33] does not offer or discuss an online AFC adaptation (R1).

The Jaccard distance is a string metric that measures the unweighted disagreement on alarm sets in two alarm subsequences. It is also applicable for AFC [5], [34].

3) ALARM SEQUENCE-BASED METHODS

Many alarm sequence-based AFC approaches align the activations of alarms in two or more alarm subsequences [5]. An example of this is the modification of the Smith-Waterman algorithm, a local sequence alignment technique initially developed for comparing genetic sequences in bioinformatics, in [35]. The modifications allow for a swapped order of alarm activations when the affected alarms are in close temporal proximity. The modified Smith-Waterman algorithm is then employed to compute the pairwise distances between an alarm subsequence to be classified and a set of historical subsequences. Subsequently, a 1NN classifier is used to determine the closest subsequence class.

One limitation of the modified Smith-Waterman algorithm in [35] is that the integration of a penalty for variations in the number of alarm activations among subsequences reduces the robustness of this method against irrelevant alarms [13]. Moreover, the calculation of the pairwise alarm subsequence distance is associated with a significant computational cost [5]. Additionally, if the historical subsequence that is nearest to the subsequence that needs to be classified was previously labeled incorrectly, the class predicted by the 1NN classifier would also be inaccurate. To solve some of these limitations, [36] and [37] extended the method proposed in [35] using a pattern extraction technique that generates a subsequence template for each class.

An online variant of the modified Smith-Waterman AFC method was proposed in [17], introducing incremental alignment with an extension strategy to minimize recalculations. However, this method may fail to consider important points of divergence where the patterns of alarms in the historical alarm subsequences and the new one start to differ, such as when there is an escalation in the underlying abnormal situation or when operators intervene effectively (R1). Subsequently, the method generates a dynamic ranking that displays historical subsequences that are most similar to the evolving one. This

allows for the presentation of subsequences that originate from different disturbances if they temporarily correspond to the new one ($R2$). However, this insight might be concealed if repetitive alarm subsequences from the same class dominate the list of rankings.

Another sequence-based AFC method, initially proposed in [38] and extended in [39], utilizes “exponentially attenuated components” (EAC) for alarm data representation, prioritizing early alarms while preserving chronological order. EAC feature vectors are determined by calculating the time distance from each alarm’s first activation to the subsequence’s start, applying exponentially attenuated weights via an attenuation coefficient reflecting process dynamics. A k-d tree facilitates efficient nearest neighbor searches. For new alarm subsequence classification, [38] employs a 1NN classifier using k-d tree distances, e.g., utilizing the L1 or L2 norm, while [39] uses Gaussian mixture models for class probability estimation.

An advantage of the approach proposed in [38] is that incorporating the Euclidean distance into the EAC-weighted relative time distances between alarm activations can compensate for variations in the alarm order when the activations are in proximity. Nevertheless, a limitation of EAC’s feature vectors in both [38] and [39] is their potential lack of tolerance for irrelevant alarms that arise early and hence carry significant weights. Moreover, although EAC feature vectors are updated incrementally ($R1$), the method’s dependence on initial alarm activations and its emphasis on the most likely class, without the ability to handle multi-labels, may restrict its ability to adapt to temporary class ambiguities and high uncertainty ($R2$).

Another AFC method based on alarm sequences was proposed in [18]. This method builds upon a concept presented in [40] and utilizes a “modified bag-of-words” (MBW) approach, a vectorization concept from natural language processing, combined with “term frequency-inverse document frequency” (TF-IDF). Both MBW and TF-IDF are employed to delineate the distinctive dynamics of a particular alarm subsequence within a feature vector, which emphasizes the early activation of alarms, similar to the methods proposed in [38] and [39]. Each feature describes a specific alarm variable and includes three weighting terms: the term frequency, the inverse document frequency, and the time weight. The resulting vectors from historical alarm subsequences are then used to train binary logistic regression classifiers for each alarm subsequence class, employing a one-versus-rest approach. If a new alarm subsequence arises, the feature vector is continuously updated with each alarm and then fed into the trained classifiers to calculate the probabilities of each class.

Because the transformation stages of the method proposed in [18] and the ones in [38] and [39] are similar, some of EAC’s properties apply, such as its tolerance for alarm order variations. Nonetheless, the method in [18] may be sensitive to irrelevant alarm activations and assign them a high

IDF weight when the corresponding alarm variable is rarely activated [13]. Moreover, because [18] employs an incremental approach triggered by alarms, it may not update class probabilities if a significant alarm that consistently activates in historical subsequences does not activate in the ongoing alarm subsequence being classified ($R1$). Furthermore, it lacks the capability to carry out multi-label classification in cases where classes cannot be distinguished with high confidence ($R2$).

In a recent study, a novel alarm sequence-based AFC method was proposed in [41]. Initially, for each alarm variable and alarm subsequence class, this approach calculates the ratio of alarm subsequences within the class that exhibit alarm activations for the particular alarm variable. It then filters out less characteristic or irrelevant alarm variables, using a minimum proportion threshold. Subsequently, the “conditional probability distribution functions” (CPDFs) of the relative alarm activation times are calculated for the selected alarm variables in an alarm subsequence class using the kernel density estimation approach with a Gaussian kernel, which is selected to smooth the probability distribution. Following that, a probabilistic fixed-order alarm subsequence matrix is computed for each historical subsequence. This matrix is based on the maximum probability of an alarm variable occurring in the associated alarm subsequence class, provided that the alarm is present in the subsequence. This matrix efficiently represents the probabilistic temporal patterns of alarms in various subsequence classes and is used as input for a “convolutional neural network” (CNN), a deep learning model commonly employed in computer vision.

Moreover, in order to make the method suggested in [41] suitable for online AFC, a technique is described to create a modified training dataset. Each historical alarm subsequence is divided into smaller segments using an expanding window technique and a fixed step size, with each segment starting at the beginning of the subsequence and continuing until a specific point, resulting in a training dataset with alarm subsequences of varying lengths. The modified training dataset is then combined with the original class labels to train a single CNN model. This feature enables the model to effectively predict known alarm subsequence classes over time, both early and incrementally ($R1$). However, it might also limit the model’s ability to learn locally meaningful patterns and raises the training complexity, particularly because the dimensions of the calculated matrices are based on the complete set of alarms defined for a plant, which may be thousands.

An inherent advantage of the method proposed in [41] is that the utilized CPDFs embed the frequency of alarm activation across all subsequences within a class as well as the level of ambiguity in the relative timing of activation. This enables it to exhibit tolerance towards swapped alarm activation orders and irrelevant alarms. However, focusing only on the first activated alarm may restrict the method’s

ability to learn all relevant alarm patterns. Moreover, the online AFC approach in [41] is unable to address temporary class ambiguities in evolving alarm subsequences. This is because the final layer of the CNN predicts only one most likely class, even if different classes may be similar ($R2$).

Alternative sequence-based AFC methods utilize different approaches to classify an alarm subsequence based on the order of activated alarms. These methods include the “basic local alignment search tool” (BLAST) [42], the “match-based accelerated alignment” (MAA) [43], the modified “closed fast sequence mining algorithm” (CloFAST) [44], the Needleman-Wunsch algorithm [45], [46], the Word2Vec algorithm derived from natural language processing for contextual alarm dependency learning [19], and “recurrent neural networks” (RNN) based on “long short-term memory” (LSTM) units [47].

4) ALARM SERIES-BASED METHODS

In [5], it was proposed to utilize alarm series instead of sequences for AFC, leveraging their ability to capture inherent dynamics more effectively, where the temporal order of alarms is less relevant. The AFC method in [5] interprets alarm series as multivariate time series, transforming each historical alarm subsequence into an “alarm coactivation matrix” (ACM). This matrix quantifies the simultaneous activity of alarm variable pairs, training a “support vector machine” (SVM) classifier. According to [5], a single classifier is trained using the complete duration of the historical alarm subsequences as input. Whenever a new subsequence appears online, the ACM is updated incrementally and the SVM classifier determines the most probable class each time a new alarm is activated.

One limitation of the method proposed in [5] arises from the presentation of the alarm series through alarm coactivations. The pairwise similarities in an ACM obscure important detail about the dynamics of an alarm subsequence, including the approximate time intervals during which alarms are active and the total duration of the situation. Disregarding these dynamics in the AFC may lead to a more ambiguous representation of alarm subsequences. Moreover, as the model is trained using all the coactivation dynamics from historical alarm subsequences, it is possible that it learns to differentiate between classes by focusing on distinctive dynamics closer to the end of the respective subsequences. This could result in initially inaccurate predictions ($R1$), particularly if temporary class ambiguity is prevalent ($R2$).

In our recent work [48], we proposed a novel alarm series-based online AFC method. This method utilizes a multivariate time series classification approach outlined in [49] and previously employed for offline AFC in our earlier study [12]. Our method combines a convolutional kernel-based transformation applied to both the binary alarm series and their first-order differential representations, capturing the occurrences of alarm activations and deactivations.

By employing various combinations of alarm variable sets, kernel configurations, and dilation factors, the method efficiently encapsulates the dynamics of alarm series in a comprehensive set of convolution outputs. These outputs are then processed to generate feature vectors for each alarm subsequence using multiple pooling operators, such as the proportion and mean of positive values, which are subsequently utilized to train an ensemble of linear ridge regression classifiers. For new alarm subsequences, our method applies transformation before classification by the ensemble, which selects the subsequence class of highest probability.

For online AFC, [48] employs an expanding window approach and an ensemble of subsequent classifiers, each tailored to a specific time segment of the underlying alarm subsequences. Our evaluation in [12] and [48] demonstrated the superiority of CASIM compared to methods previously proposed in [5], [18], [33], [35], [38], and [39]. As CASIM incorporates both alarm series and the sequence of alarm activations and deactivations as input, it is capable of learning more significant patterns. Nevertheless, CASIM lacks the capability to differentiate temporarily ambiguous classes ($R2$). However, it has demonstrated a rather rapid ability to differentiate classes once the information becomes accessible to the model. This is due to the utilization of the expanding window approach, together with the ensemble of locally specialized models, which enables CASIM to concentrate on local features and make incremental and early predictions ($R1$).

5) CLASSIFICATION UNCERTAINTY

Addressing AFC uncertainty and temporary class ambiguities by presenting a list of the most likely classes or using distance or probability thresholds presents considerable challenges. Firstly, opting for a fixed number of top classes to be displayed at every instance results in a presentation that might fail to adapt to the varying degrees of ambiguity. Secondly, setting a distance or probability threshold for class inclusion requires calibration to ensure the true class is consistently captured within the suggested set [23], [50], [51]. Lastly, relying on posterior class probabilities from ML models introduces biases linked to class distribution within the training set. An imbalance can skew predictions towards overrepresented classes, despite the actual indistinguishability of the classes involved. Conversely, in well-balanced scenarios, the probability distribution across temporarily ambiguous classes may not accurately reflect their true likelihood, especially when multiple classes exhibit identical characteristics [23].

In conclusion, the described AFC methods show deficits in fulfilling both the requirements $R1$ and $R2$. This justifies the proposal of a novel method-agnostic approach, enabling the existing AFC methods to effectively deal with ambiguity and uncertainty, and continuously update the classification as the alarm subsequence progresses over time.

B. CONFORMAL PREDICTION

1) OVERVIEW

CP, also known as conformal inference [50], [51], offers a framework for generating statistically valid prediction sets, ensuring that predictions include the true outcome with a specified level of confidence, without relying on distributional assumptions [23], [51], [52]. In fact, CP can be seen as a method for taking any heuristic notion of uncertainty from a supervised prediction model and converting it to a statistically rigorous one [23], [51]. A detailed introduction to CP is given both in [23] and [51].

Typically, CP is applied in domains requiring robust uncertainty quantification to support critical decision-making processes [23], [50], [51]. In medical diagnostics, CP is used to generate prediction sets that highlight possible diagnoses, such as those derived from X-ray images, allowing medical practitioners to anticipate multiple possible outcomes, which is particularly beneficial when it is not possible to definitively exclude critical medical conditions [53]. Similarly, in financial forecasting, CP provides a range of potential market movements, helping analysts account for uncertainty and prepare strategies for different scenarios [54]. Beyond these domains, CP has been applied in autonomous driving to assist in online decision-making by providing a range of potential future states of dynamic environments, such as intersections with pedestrians, using sensor data streams. In this context, having access to multiple plausible predictions allows the system to avoid dangerous maneuvers under uncertain conditions and employ more cautious strategies [55]. The aforementioned applications illustrate that CP can be applied to both classification and regression problems [23], [51].

The relevant literature primarily identifies two types of CP: transductive CP and inductive CP [23], [51]. Transductive CP, or full CP [23], [51], [56], [57], operates directly on individual test instances, making predictions based on both the training data and the specific sample under consideration. It computes nonconformity scores for each possible label of the test instance by repeatedly retraining the model. This is done by assuming that the test instance belongs to the tested label and temporarily including it into the training set. The prediction set then consists of all labels that have a nonconformity score at or below a selected threshold. This approach is efficient on the available training data and allows for accurate prediction sets [23], [51], [56], [57]. However, the drawback of this approach is its high computational expense, since it necessitates the need to retrain the model multiple times for every new test instance [23], [51], making it less practical for online applications, such as online AFC.

Inductive CP, or split CP [23], [51], on the other hand, divides the available data into two subsets: one for training and the other for calibration. The prediction model is trained only once on the training set, and the nonconformity scores are computed using the calibration set, resulting in a faster prediction in comparison to transductive CP [23], [51]. The trade-off in this case is that a substantial quantity

of labeled data is not available for training and must be retained for calibration, potentially negatively impacting the overall performance of the prediction model. Nevertheless, for applications requiring prompt decisions, such as online AFC, inductive CP provides statistically reliable uncertainty estimates without the computational burden of transductive CP [23], [51], [58], [59]. Inductive CP was initially presented for regression in [58] and for classification in [59].

2) INDUCTIVE CONFORMAL PREDICTION

Inductive CP is particularly relevant for this study as it offers a more computationally efficient alternative to traditional transductive CP, making it highly suitable for online AFC in industrial contexts.

In the context of multi-class classification, inductive CP utilizes a pre-trained classification model alongside a calibration dataset. The latter contains class instances that were not exposed to the model during its training phase. After processing the calibration data, the model outputs predictions, each accompanied by a score function, which quantitatively assesses the model's confidence in associating the prediction with each possible class [23], [51], [59]. These scores might take the form of posterior class probabilities, commonly utilized in ML techniques, or distance measures, as seen in sequence alignment methods.

To calibrate the prediction output, the model's scores for the true class—based on the known ground-truth labels of the calibration set—are aggregated. This aggregation can occur across all classes or be class-specific, leading to the construction of a score distribution for the true class. Subsequently, an error rate parameter, denoted by $\alpha \in [0, 1]$, is employed to derive a threshold or a set of class-specific thresholds from this distribution. When classifying a new instance, the model's scores for each potential class are evaluated against these thresholds. A class is considered a plausible label if its score surpasses the corresponding threshold, culminating in a set of likely and potentially ambiguous class labels. This set is guaranteed to contain the true label with a minimum average probability of 1α ; a property known as marginal coverage, where the term coverage refers to the proportion of times the true label is included in the predicted sets over a given test dataset [23], [51], [59].

3) OTHER RELEVANT DEVELOPMENTS

Several specific methods within CP, such as adaptive prediction sets [23] and conformalized quantile regression [60], address different aspects of predictive modeling. These techniques adapt to the unique challenges of each application, from handling classification problems with dynamic prediction sets to providing calibrated uncertainty estimates in regression tasks. The resulting predictions provide more information about the uncertainty surrounding a potentially ambiguous situation compared to a standard point prediction [23].

Alternative CP methods such as cross-CP [61] as well as Jack-knife+ and CV+ [62] provide a trade-off between computational and statistical efficiency. Unlike inductive CP, these approaches use all available data for both model training and calibration, while minimizing the number of model fits, although not completely avoiding them [51].

CP also extends to handling more specialized tasks like outlier detection [63], covariate shift [64], and distribution shift [65]. For outlier detection, CP adapts its framework to identify points that deviate from the distribution of a clean dataset by assigning scores to test points and flagging those exceeding a conformal threshold as outliers [63]. CP under covariate shift addresses scenarios when the distribution of the test data deviates from the calibration data, although the relationship between features and labels remains constant. In such cases, the weights of calibration points are adjusted according to their probability in the new distribution to maintain valid prediction sets [64]. Lastly, CP under distribution shift, also referred to as adaptive CP [66], [67], [68], [69], [70], [71] or online CP [72], addresses the issue of gradually changing data distributions. In this scenario, the relationships between features and labels change as well. By assigning greater importance to recent data, adaptive CP is able to adjust prediction sets as conditions evolve [65], [66], [67], [68], [69], [70], [71], [72], particularly in time series contexts [68], [69]. While adaptive CP addresses dynamic predictions in an online manner, its primary emphasis is on adjusting to variations in the incoming data stream relative to the calibration dataset, rather than regarding temporary class ambiguities that are inherently uncertain for a limited amount of time.

4) SUMMARY

Despite its method-agnostic properties and broad applicability, CP has not yet been directly applied to address uncertainty in industrial alarm management. Moreover, CP currently does not offer solutions to resolve temporary class ambiguity, where higher levels of uncertainty can be limited in duration.

C. EARLY TIME SERIES CLASSIFICATION

1) OVERVIEW

ETSC is a field that specifically deals with the issue of temporary class ambiguities found in evolving time series data [28]. These methods aim to optimize the balance between early classification and accuracy, i.e., increasing the amount of data observed usually improves accuracy but can delay classification [73]. This balance can be critical in scenarios where a decisive, yet timely, identification can significantly influence decision-making processes, such as early diagnosis in hospital emergency rooms [74], [75], earthquake early warning [76], and industrial control rooms [77]. Another related, but more general, field of study is “ML-based early decision making” (ML-EDM) [78].

Nevertheless, in [79], the authors argue that ETSC is flawed, primarily due to the focus on ambiguous early time series segments, which can lead to inaccurate predictions.

Furthermore, they propose that in most use cases, there is no necessity to compromise accuracy and timeliness, as delaying until slightly more data is available can yield adequate information for more precise predictions without compromising much decision-time. This strategy, however, would not provide any decision-support during periods of temporary class ambiguity. Moreover, the authors in [79] criticize the use of z-normalization in ETSC, noting that it can artificially boost results by peeking into future data. However, we argue that in the context of AFC, this concern is mitigated because alarm series are naturally binary, eliminating the need for z-normalization and making ETSC more applicable.

By contrast to [79], the authors in [80] advocate for ETSC, highlighting its importance in time-critical scenarios where early decisions can prevent costly delays. They claim that ETSC can achieve an effective balance between timeliness and accuracy. They also highlight that ETSC offers the principles and procedures that can determine when uncertainties like temporary class ambiguities have been resolved. Nevertheless, ETSC does not offer assistance as long as the temporary ambiguity persists.

Reference [81] introduces a taxonomy to categorize ETSC methods with a specific emphasis on those that employ a separated architecture. The latter specifies an architectural design that includes a specific trigger function, responsible for determining the point at which sufficient data has been observed, and a classification function, responsible for generating class predictions. An alternative is an end-to-end architecture in which both components are integrated, as exemplified in [82].

In [81], one dimension refers to the classification strategy used for the classification function, which can be implemented either by employing a collection of classifiers, each specifically designed for a certain time segment, or by using a single classifier for all data points. Another dimension pertains to the optimization strategy employed during the training of the trigger function. This strategy can be either cost-informed, which considers the costs associated with misclassification and delay, or cost-uninformed, which only relies on confidence-thresholds or static rules such as a pre-determined time that dictate when classification should be performed [81]. A third dimension describes the trigger function’s inference properties. This includes a confidence-based trigger that considers only a single instance or sequence of classification confidences, a cost-informed approach, or an anticipation-based approach. The latter acquires cost patterns during the training process, allowing for the prediction of how the cost will evolve during inference, provided that the test sample is drawn from the same distribution as the training samples [81].

2) EXISTING METHODS

A comprehensive analysis of existing ETSC methods was recently presented in [81], along with a thorough assessment of these methods in relation to one another and benchmarks

such as classification “as soon as possible” (ASAP) and classification “as late as possible” (ALAP). Supplementary literature reviews and analyses can be found in [28] and [83].

For example, the “two-tier early and accurate series classifier” (TEASER) method, introduced in [84], utilizes an ensemble of primary classifiers. Each classifier is trained on a distinct segment of the time series data and used to generate probability scores for each class. These scores are then fed into a secondary classifier that determines if sufficient dynamics of a time series have been observed to confidently output the class label predicted by the respective primary classifier. The cost-informed and confidence-based approach used in TEASER has been shown to outperform other state-of-the-art ETSC approaches in terms of both accuracy and earliness in [28], [84], [85], and [86].

The “calibrated early time series classifier” (CALIMERA), proposed in [86], is another cost-informed ETSC method that utilizes an ensemble of classifiers trained on predetermined lengths of the time series data. CALIMERA is an anticipation-based ETSC approach that utilizes classifier calibration to determine the expected classification performance. This information is used to make decisions on when to wait or stop, with the goal of minimizing a cost function. Alternative ETSC approaches utilize LSTM for classification and leverage its hidden state for determining stopping criteria [87]. They also incorporate adaptive convolutional kernels and random truncation-based data augmentation techniques to adjust to varying characteristics and lengths of dynamic time series [88].

Several other ETSC methods are described in the literature. For example, the anticipation-based method developed in [89] initiates a classification when the cost at a specific time step is lower than the predicted cost at any subsequent time step. Within a confidence-based method proposed in [90], a stopping rule is activated by a combination of classification confidence estimates and a dynamic delay measure. Another confidence-based approach in [91] aggregates the classification confidence up to a specific time point and passes the most probable class to the human operator if a predetermined threshold is exceeded. Lastly, another confidence-based ETSC method in [92] establishes a set of secure timestamps by considering the performance of the classifier on historical data. This ensures that predictions are only generated if both any of the timestamps is reached and a threshold for class probability is satisfied.

3) SUMMARY

To summarize, ETSC provides useful mechanisms for training classifier ensembles that specialize in certain time segments and effectively trigger classification at optimal points, balancing timeliness and accuracy [80]. However, the employed trigger functions do not handle phases of temporary class ambiguity, as they cannot provide predictions before or after a classification decision [79]. Furthermore, ETSC’s classification functions are not able to generate sets of

ambiguous classes when uncertainty persists. In conclusion, although ETSC provides useful techniques for classifying evolving time series, it falls short of addressing the challenge of managing uncertainty during periods of temporary class ambiguity, which is essential for online AFC.

D. FINDINGS SUMMARY

Our analysis shows that existing AFC methods are effective at differentiating various characteristics of alarm data, however, their capabilities are limited when encountering temporary class ambiguity and high AFC uncertainty. CP introduces a framework for managing this ambiguity by predicting a set of potential classes based on probabilistic confidence levels, yet it falls short of addressing the dynamic, evolving nature of alarm floods. On the other hand, ETSC can adapt to the evolution of time series data, offering timely and accurate classification, but lacks mechanisms to explicitly deal with the early stages of ambiguity and uncertainty before clear class distinctions emerge.

This research gap justifies the development of a novel method that integrates both CP and ETSC, allowing the system to manage the dynamic nature of online AFC while providing operators with early decision support, even during periods of temporary class ambiguity.

III. PROPOSED APPROACH

This section details the methodology of our proposed approach. It starts with an overview of the conceptual framework integrating CP and ETSC into AFC in Subsection III-A. In Subsection III-B, we provide a detailed description of the steps involved in both the calibration and inference phases of our approach. Finally, we discuss the advantages and limitations of our method in Subsection III-C.

A. OVERVIEW OF THE PROPOSED APPROACH

Considering the findings in Subsection II-D, we propose a novel synergistic approach that integrates elements of both CP and ETSC into AFC. This integration aims to harness CP’s advantages in handling uncertainty and ETSC’s proficiency in managing evolving time series data, thereby offering a comprehensive solution to the challenge of temporary class ambiguity and uncertainty in AFC.

Our proposed “conformal and early AFC” (ConE-AFC) method is designed to be method-agnostic, allowing it to seamlessly integrate as an additional, complementary stage within existing AFC methods and implementations, which traditionally comprises transformation, classification, and novelty detection. Fig. 2 shows the “formalized process description” [93] of the proposed ConE-AFC method. The method consists of two phases: calibration using historical data (Fig. 2(a)) and inference with a new alarm subsequence online (Fig. 2(b)). This methodology is consistent with conventional CP-based approaches [23], [50], [52]. Each phase consists of process operators (green rectangles) and the corresponding processed and generated information (blue hexagons).

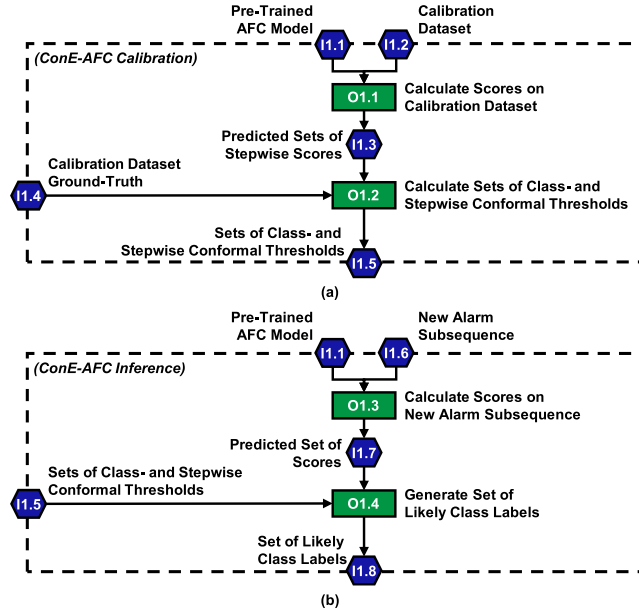


FIGURE 2. “Formalized process description” of the proposed “conformal and early AFC” (ConE-AFC) method. (a) Calibration on historical data. (b) Inference on online data.

In the proposed ConE-AFC method, the calibration phase depicted in Fig. 2(a) involves utilizing a pre-trained AFC model (I1.1) and a calibration dataset (I1.2) to initially compute the prediction scores (I1.3), such as class probabilities or distances, for each alarm subsequence in the calibration dataset during step O1.1. The scores are computed for each of the known historical classes, as well as for a sequence of time intervals where only partial data is provided to the AFC model, while the remaining data is concealed to imitate the evolving characteristics of alarm subsequences. These predicted sets of stepwise scores are subsequently combined with the ground-truth labels of the calibration dataset (I1.4) in step O1.2 to compute sets of class- and stepwise conformal thresholds (I1.5). This stepwise calibration ensures that our method adapts dynamically to the evolving nature of alarm subsequences while effectively addressing the inherent ambiguity and resulting uncertainty, thus potentially enhancing the reliability of AFC as a tool for human experts.

In the inference phase depicted in Fig. 2(b) of our proposed method, we utilize the same pre-trained AFC model (I1.1) as in the previous phase. We employ this model to classify a new alarm subsequence (I1.6) that has not been observed before, in step O1.3. The obtained scores (I1.7) are subsequently compared to the calibrated thresholds (I1.4) to determine the class labels of likely classes (I1.8) that should be given to the operator following the inference phase.

B. DETAILS OF THE PROPOSED APPROACH

The ConE-AFC method starts with the calibration phase in Fig. 2(a) and step O1.1, which involves utilizing an AFC model (I1.1) pre-trained on historical alarm data and capable of providing a score for each known alarm subsequence class.

These scores, which can either be posterior class probabilities or distance measures, are computed for each class $k \in K$, where K is the set of known classes.

In addition, a calibration dataset X (I1.2) is meticulously constructed from a subset of historical alarm subsequences $x \in X$ that the model has not encountered during its training phase [23]. The marginal coverage guarantee ensures that the coverage will be at least $1 - \alpha$ on average across the calibration set’s randomness, independent of the size n_k for each class $k \in K$ within the dataset. However, the coverage on an infinite test dataset with a fixed calibration dataset may not be exactly $1 - \alpha$. According to formulas given in [23] and [94], we can precisely calculate a sufficiently large n_c for each class c in the calibration dataset to manage these fluctuations in coverage, i.e., to achieve a coverage of $1 - \alpha \pm \epsilon$ with a confidence of $1 - \delta$. For example, when targeting $\alpha = \delta = \epsilon = 0.1$, n_k should be 22 for each $k \in K$ [23].

Fig. 3 illustrates the interpretation of these three parameters in the application of our proposed ConE-AFC method to an infinite number of test samples. In other words, the expected coverage for classifying any given alarm subsequence instance is between 0.8 ($1 - \alpha - \epsilon$) and 1.0 ($1 - \alpha + \epsilon$) with a 90% confidence interval ($1 - \delta$).

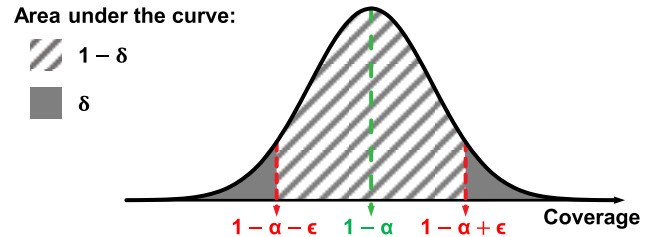


FIGURE 3. The distribution of coverage is illustrated by an infinite test set, indicating the properties associated with the parameters α , δ , and ϵ .

Nevertheless, this definition may prove challenging for a human operator lacking expertise in ML or statistical fundamentals to comprehend. Therefore, we offer a singular metric to enhance comprehension of the prediction’s reliability. As we are also interested about what is located to the right of the selected confidence interval, we can establish the anticipated probability that the actual class falls within the predicted set of potential labels, i.e., the expected coverage $P(C)$, as follows:

$$P(C) = (1 - 0.5\delta) \times (1 - \alpha - \epsilon). \quad (1)$$

This definition is valid if n_k is the same for all $k \in K$. Alternatively, the expected coverage is conditional upon the actual class label of the alarm subsequence that is being classified. In our given example, the calculation of $P(C)$ results in 0.76, indicating that the operator can expect a minimum coverage of 76%. However, assessing the usability and interpretability of the expected coverage provided in (1) is not within the scope of this technical paper.

Following TEASER [84], our proposed ConE-AFC method employs an expanding window technique for

segmenting the historical alarm subsequences in the calibration dataset into incremental pre-fixed windows. This ensures that the method remains adaptable to the dynamic nature of present uncertainties, e.g., due to temporary class ambiguity, allowing for more accurate prediction sets in the early stages of evolving alarm dynamics by focusing on smaller subsequence segments during calibration.

The expanding window technique is shown in Fig. 4, with $T(t_i)$ denoting the window that begins at zero and extends to the endpoint at the i -th time step $t_i = i \times \omega$, where ω represents the selected pre-fixed interval length. Alarm subsequences in the calibration dataset that are shorter than a given window are padded with zeros at the end to ensure compatibility with any AFC model [12], [48]. From this, we can generate the subset $X_i \subseteq X$, where each $x_i \in X_i$ corresponds to the i -th segment of the subsequence x bounded by window $T(t_i)$.

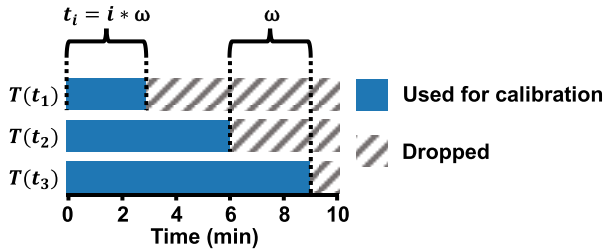


FIGURE 4. Concept of expanding windows to segment an alarm subsequence for calibrating the proposed ConE-AFC method. With window T , step t , and interval length ω .

The aforementioned concepts are utilized in O1.1, along with the pre-trained AFC model (I1.1) and the alarm subsequences $x \in X$ in the calibration dataset (I1.2). Consequently, we compute the stepwise score sets $S_i \subseteq S$ (I1.3) for each window $T(t_i)$ using the corresponding X_i . These score sets represent the prediction scores $s_k(x_i)$ for each alarm subsequence $x_i \in X_i$ and class $k \in K$. For example, in case of posterior class probabilities, $s_k(x_i)$ can be written as posterior probability $P(k | x_i)$ of class k given x_i .

In the next step of our proposed ConE-AFC method, O1.2, we employ the stepwise score sets $S_i \subseteq S$ (I1.3) and the corresponding ground-truth class labels $y(x) \in Y(X)$ (I1.4) of the alarm subsequences $x \in X$ to compute the class- and stepwise conformal thresholds $\mu_{i,k}$ (I1.5) for each window $T(t_i)$ and class k . Following [23], we first define the subset $X_{i,k} \subseteq X_i$ as all alarm subsequences $x_i \in X_i$ for which the true label $y(x) = k$, i.e., $X_{i,k} = \{x_i \in X_i | y(x) = k\}$, regardless of whether the model accurately predicted the class or not. Next, we obtain all scores for each subset $X_{i,k}$ that corresponds to its label k , i.e., $S_{i,k} \subseteq S_i$ with $S_{i,k} = \{s_k(x_i) \in S_i | \forall x_{i,k} \in X_{i,k}\}$. The scores in $S_{i,k}$ are then sorted in descending order when posterior class probabilities are used to obtain a sequence $\hat{S}_{i,k}$. From this, we can calculate $\mu_{i,k}$ as follows (following [23]):

$$\mu_{i,k} = \hat{S}_{i,k} \left[\lceil (1 \times \alpha) \times |\hat{S}_{i,k}| \rceil \right], \quad (2)$$

where $|\hat{S}_{i,k}|$ is the total number of scores in $\hat{S}_{i,k}$, $\lceil \cdot \rceil$ is the ceiling function, which rounds up to the nearest integer, and $\hat{S}_{i,k}[\cdot]$ denotes the score at a particular index in $\hat{S}_{i,k}$.

Formula (2) guarantees that a minimum proportion of 1α of the alarm subsequences $x_{i,k} \in X_{i,k}$ have a score $s_k(x_i)$ for the true class label k that exceeds the threshold $\mu_{i,k}$. Fig. 5 displays a synthetic distribution of posterior class probabilities and the corresponding descending cumulative percentage. This diagram visually illustrates how the ratio 1α can be utilized to derive the conformal threshold μ .

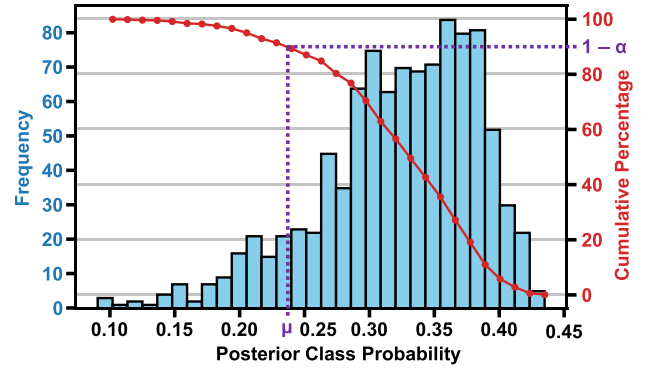


FIGURE 5. Histogram of synthetic data illustrating frequency distribution of posterior class probabilities for class c , accompanied by a cumulative percentage curve starting at 100% and decreasing progressively. The conformal threshold μ is determined by calculating the posterior probability at which a fraction of $1 - \alpha$ samples have a probability value greater than μ .

If distance measures are employed for scoring, the sequence $\hat{S}_{i,k}$ must be arranged in ascending order. The conformal threshold $\mu_{i,k}$ can be determined by the following calculation:

$$\mu_{i,k} = \hat{S}_{i,k} \left[\lfloor (1 \times \alpha) \times |\hat{S}_{i,k}| \rfloor \right], \quad (3)$$

where $\lfloor \cdot \rfloor$ is the floor function, which rounds down to the nearest integer. Formula (2) ensures that at least a proportion of 1α of the alarm subsequences $x_{i,k} \in X_{i,k}$ have a score $s_k(x_i)$ for the true class label k that is lower than the threshold $\mu_{i,k}$.

The calculation of the conformal thresholds $\mu_{i,k}$ (I1.5) for each window $T(t_i)$ and each class $k \in K$ concludes the calibration of our ConE-AFC method shown in Fig. 2(a). The calibration phase of our method is further described in algorithmic form in Algorithm 22. Subsequently, upon the detection of a novel alarm subsequence $\tilde{x} \notin X$, the second stage of our approach is initiated, namely the online inference stage, as depicted in Fig 2(b).

The identification of $\tilde{x}$ can be accomplished through various techniques, such as when the alarm activation rate surpasses a specific threshold within a 10-minute timeframe or employing outliers in the distances between alarm activations and

deactivations [16]. The outcome is an alarm subsequence $\tilde{x}$ that undergoes evolution and consequently expands over time. To ensure comparability, we utilize the same concept of expanding windows as we do for the alarm subsequences

Algorithm 1 ConE-AFC Calibration Phase

Input:
M: Pre-trained AFC model
X: Calibration dataset
Y: Ground-truth labels for X
K: Set of known classes
 α : Error rate parameter
 ω : Interval length for expanding windows
L: Length of the subsequences in X

Initialize Expanding Windows:

```

1  Define the set of expanding windows
    $T = \{T(t_i) | t_i = i \times \omega, i = 1, 2, \dots, \lfloor L/\omega \rfloor\}$ .
Score Computation:
2  for each window  $T(t_i) \in T$  do:
3      Initialize an empty list  $S_i$  for window  $T(t_i)$ .
4      Generate the subset  $X_i \subseteq X$ , where each  $x_i \in X_i$  corresponds to  $x$  segmented using  $T(t_i)$ .
5      for each subsequence  $x_i \in X_i$  do:
6          Compute the score  $s_k(x_i)$  for each class  $k \in K$  using the model  $M$ .
7          Append the scores  $s_k(x_i)$  to the list  $S_i$ .
8      end
9  end
Threshold Computation:
10 for each window  $T(t_i) \in T$  do:
11     for each class  $k \in K$  do:
12         Extract the subset  $X_{i,k} \subseteq X_i$  where the true label  $y(x) = k$  from  $Y$ .
13         Extract the corresponding  $S_{i,k} \subseteq S_i$ .
14         Determine whether  $M$  returns distances or probabilities:
15         if  $M$  returns probabilities, then sort  $S_{i,k}$  in descending order to obtain  $\hat{S}_{i,k}$ .
16         else if  $M$  returns distances, then sort  $S_{i,k}$  in ascending order to obtain  $\hat{S}_{i,k}$ .
17         Calculate the conformal thresholds  $\mu_{i,k}$ :
18         if  $M$  returns probabilities, then use (2).
19         else if  $M$  returns distances, then use (3).
20     end
21 end
Return:
22  $\mu_{i,k}$  : Class- and stepwise conformal thresholds  $\{\mu_{i,k} | T(t_i) \in T, k \in K\}$ 

```

$x \in X$ in the calibration dataset. In other words, $\tilde{x}_i$ represents the i -th segment of $\tilde{x}$ that is defined by the window $T(t_i)$.

First, in O1.3, the pre-trained AFC model (I1.1) is employed together with the currently available $\tilde{x}_i$ (I1.6) to predict the class-wise scores $\tilde{S}_i = \{s_k(\tilde{x}_i) | \forall k \in K\}$ (I1.7).

Subsequently, in O1.4, the aforementioned scores are used in conjunction with the conformal thresholds $\mu_{i,k}$ (I1.5) for the i -th segment and each class $k \in K$ to calculate the corresponding prediction set $\tilde{Y}_i^{\text{pred}}$ (I1.8), encompassing all possible alarm subsequence classes that are to be taken into account. In the case of AFC models that incorporate posterior class probabilities, the calculation of $\tilde{Y}_i^{\text{pred}}$ can be expressed as follows (following [23]):

$$\tilde{Y}_i^{\text{pred}} = \{k \in K | s_k(\tilde{x}_i) \geq \mu_{i,k}\}. \quad (4)$$

Similarly, for AFC models that generate distance measures, wherein smaller distances indicate greater class agreement, the calculation can be performed as follows:

$$\tilde{Y}_i^{\text{pred}} = \{k \in K | s_k(\tilde{x}_i) \leq \mu_{i,k}\}. \quad (5)$$

Finally, the human operator can be provided with the resultant $\tilde{Y}_i^{\text{pred}}$ and expected coverage $P(C)$ utilizing (1) in order to offer an uncertainty quantification that can aid in the decision-making process. As soon as new information becomes available and the ongoing subsequence $\tilde{x}$ increases, i.e., the $(i+1)$ -th segment becomes available, a new prediction $\tilde{Y}_{i+1}^{\text{pred}}$ is made. The inference phase of our method is further described in algorithmic form in Algorithm 14.

C. LIMITATIONS AND ADVANTAGES OF THE PROPOSED APPROACH

One of the limitations of the ConE-AFC method is the assumption that the new alarm subsequence $\tilde{x}$ is derived from the same distribution as the subsequences in the calibration dataset. Nevertheless, it is important to note that this assertion may not hold true, thus rendering statistical confidence measures and guarantees potentially invalid [23]. However, we argue that AFC models commonly include a novelty detection stage that aims to identify alarm subsequences that may come from a previously unobserved class, i.e., subsequences that are “out-of-distribution” (OOD). Consequently, any final prediction provided to the operator should include both the results obtained from the novelty detection stage’s OOD detection and the ConE-AFC method’s CP. The objective of this study is to provide a proof of concept for the utilization of CP and ETSC to address uncertainty in online AFC. Hence, the evaluation of a combined approach involving a novelty detection stage and the ConE-AFC method falls outside the scope of this study.

Another limitation of the ConE-AFC method stems from its reliance on extensive data to guarantee a specified coverage. For instance, [23] indicates that achieving a $P(C)$ of 0.85, with $\alpha = \delta = 0.1$ and $\epsilon = 0.01$, necessitates a calibration dataset size n_k of 2491 for each class $k \in K$. This requirement potentially limits our method’s applicability in industrial scenarios where such extensive datasets are not readily available. To mitigate this limitation, we suggest exploring data augmentation techniques [95] or leveraging process simulators. These approaches can potentially expand the dataset, providing the volume of data required for CP calibration. Nevertheless, it is important to evaluate the quality

Algorithm 2 ConE-AFC Inference Phase

Input:
 M : Pre-trained AFC model
 T : set of expanding windows
 K : Set of known classes
 μ : Conformal thresholds $\{\mu_{i,k} | T(t_i) \in T, k \in K\}$
 $\tilde{x}_i$: i -th segment of the new alarm subsequence to be classified defined by the window $T(t_i)$

Score Computation:

- 1 Compute the prediction scores $\tilde{S}_i = \{s_k(\tilde{x}_i) | \forall k \in K\}$ for $\tilde{x}_i$ using M .

Threshold Comparison and Class Prediction:

- 2 Initialize an empty set $\tilde{Y}_i^{pred}$ to store the predicted classes.
- 3 Determine whether the model M returns distances or probabilities.
- 4 **if** M returns probabilities, **then**:
- 5 **for each** $k \in K$ **do**:
- 6 **if** $s_k(\tilde{x}_i) \geq \mu_{i,k}$, **then add** k to $\tilde{Y}_i^{pred}$.
- 7 **end**
- 8 **end**
- 9 **else if** M returns distances, **then**:
- 10 **for each** $k \in K$:
- 11 **if** $s_k(\tilde{x}_i) \leq \mu_{i,k}$, **then add** k to $\tilde{Y}_i^{pred}$.
- 12 **end**
- 13 **end**

Return:

- 14 $\tilde{Y}_i^{pred}$: Set of likely class labels for $\tilde{x}_i$

and validity of artificially enhanced data in order to meet the CP's criteria for statistical coverage.

On the other hand, our proposed ConE-AFC method has the advantage of being method-agnostic, i.e., it can be directly implemented on any pre-trained AFC model. The sole prerequisite is that each known class must have an assigned measure (be it posterior probabilities or distance measures), and a calibration dataset is needed. Such flexibility enhances the utility of existing AFC models by equipping them with capabilities to address uncertainty and improve classification efficacy with no extensive model reengineering. Furthermore, in response to requirement *R1* in Section II, our proposed ConE-AFC method is designed to effectively handle the evolving characteristics of alarm subsequences. By employing expanding windows from the ETSC domain, our approach ensures the incremental and timely CP of alarm subsequences as they evolve. Moreover, our method enables incremental CP even if the underlying AFC model was not trained incrementally.

One additional advantage of our approach is its ability to effectively handle temporary class ambiguity (*R2*)—a com-

mon challenge during the early stages of alarm floods [17], [19]—as well as model uncertainty. By integrating CP, our ConE-AFC method offers multi-label outputs, thus providing operators with a reliable set of likely classes along with their expected coverage. In line with [23] and [32], we argue that this multi-faceted output can be beneficial in situations where initial alarms might not distinctly point towards a single subsequence class but result in ambiguity, potentially improving operator trust in a deployed AFC model.

IV. EVALUATION AND DISCUSSION

This section focuses on assessing the performance and characteristics of five existing AFC methods from Section II when utilized as intended by their authors, and in contrast the application of our ConE-AFC approach to them, as outlined in Section III. In Subsection IV-A, the evaluation dataset is summarized, and in Subsection IV-B, suitable evaluation measures are selected. The experimental setup is described in Subsection IV-C. The evaluation results are presented and discussed in Subsection IV-D.

A. SYNTHETIC ALARM EVALUATION DATASET

Based on the systematic principles of “design science research”, evaluation involves a balance between the applicability to real-world situations and the scientific rigor of the findings [96]. This work aims to demonstrate the applicability of CP and ETSC to AFC using the proposed method ConE-AFC. A formal mathematical proof of the conformal predictor, which our method is based on, has already been provided in [23] and [94].

To carry out a controlled experiment, it is necessary to have a dataset that has an objective ground-truth and is of adequate size and diversity [96]. However, the availability of large datasets for real industrial plants is limited due to the high cost and potential risks associated with their creation, particularly in the context of abnormal situations and alarm management [97]. Furthermore, the ground-truth may not be readily accessible in such instances, as the interpretation and labeling of the dataset require the knowledge of human experts [97]. Another option is to use datasets generated from realistic simulation models of industrial plants. Conveniently, the ground-truth in a simulation dataset can be determined by the known types of the simulated disturbances [98].

However, the specific point in time at which two different disturbances become distinguishable is not clearly defined and may be influenced by random factors [98], [99]. Therefore, datasets generated using process plant simulation models may not readily provide the ground-truth information regarding temporary class ambiguities, which is needed for our evaluation. Furthermore, for a comprehensive assessment of the calibration of the ConE-AFC method, as explained in [23] and [94], it is crucial to have a substantial number of samples per class. Nevertheless, employing complex simulation models for this purpose would result in substantial expenses in terms of computational resources. To overcome these limitations, we systematically designed and generated a

synthetic evaluation dataset to replicate the features of actual alarm dynamics. This dataset includes a definitive reference for classes that have temporary ambiguity, while also being sufficiently simple to generate large sample sizes with low computational costs. In [33], a similar methodology was employed, generating a dataset consisting of synthetic alarm sets to assess the effectiveness of AFC methods.

Our synthetic dataset consists of alarm series and uses alarm probabilities for each of the implemented ten alarm variables to be active at each time step, while the sampling rate is set to 1/min, in order to simulate the behavior of realistic alarm data with temporary class ambiguities. These probabilities are dependent on the alarm subsequence class and the current simulation time of the subsequence. In order to replicate the behavior of alarm classes that initially have ambiguous characteristics but gradually develop distinct features, all five classes begin with identical probabilities assigned to the ten alarm variables. The dendrogram in Fig. 6 illustrates the precise timing and probabilities associated with each of the ten alarm variables and the five alarm subsequence classes. It is evident that each alarm subsequence class can be completely distinguished from a specific moment in time.

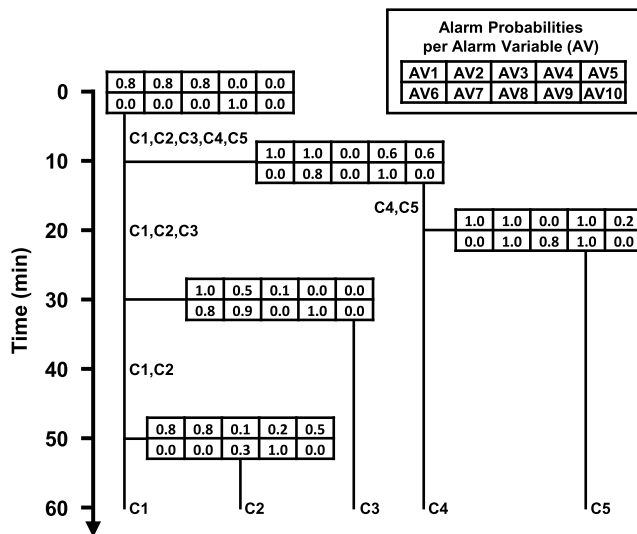


FIGURE 6. Dendrogram of the time-dependent alarm probabilities for the 10 alarm variables (AVs) and the 5 alarm subsequence classes (C1 to C5). Multiple class labels on the same branch represent temporary class ambiguity.

However, due to the probabilistic nature of the alarm subsequence class definitions, we made sure to achieve a reasonably high level of complexity for the AFC task. In addition, the alarm variable 9 is consistently active, while alarm variable 10 remains inactive at all times. This phenomenon is commonly observed in alarm systems that are not properly configured [8], [9], which adds complexity to the AFC, as the AFC model needs to handle these values that lack any useful information.

In our dataset, there are 3,750 subsequences per class, and each subsequence is generated using the probabilities

shown in Fig. 6. There are a total of 18,750 subsequences across the five classes. The dataset can be accessed publicly through the DOI hyperlink provided in the code and data availability statement at the conclusion of this paper. Fig. 7 illustrates the alarm dynamics for each of the five classes and all ten alarm variables selecting a random subsequence from the dataset for each of the five classes. It is evident that over time, the alarm dynamics become more pronounced, which makes it challenging to differentiate between classes at early time points. Furthermore, it is noticeable that only by considering a comprehensive perspective on multiple alarm variables can we obtain sufficient information to differentiate between classes.

B. PERFORMANCE METRICS

To evaluate the efficacy of our ConE-AFC approach when applied to existing AFC models, we utilize three metrics that aim to measure the accuracy of single-label predictions and the comprehensiveness of multi-label outputs.

Accuracy serves as our primary metric for evaluating the performance of the original AFC models, which produce single-label outputs. It measures the proportion of correctly predicted alarm subsequences out of the total number of subsequences in the test dataset. Mathematically, it is defined as:

$$\text{Accuracy} = \sum_{n=1}^N 1(y_n = \hat{y}_n) / N, \quad (6)$$

where N is the total number of alarm subsequences in the test dataset, y_n is the true class of the n -th alarm subsequence, $\hat{y}_{n,i}$ is the predicted class of the n -th alarm subsequence, and $1(\cdot)$ is the indicator function that is 1 if its argument is true, and 0 otherwise.

For our ConE-AFC method, which yields multi-label outputs, coverage quantifies the extent to which the true subsequence classes are included within the predicted sets. It is calculated as (following [23]):

$$\text{Coverage} = \sum_{n=1}^N |y_n \cap \hat{Y}_n| / N, \quad (7)$$

where $\hat{Y}_n$ is the set of predicted classes for the n -th alarm subsequence, and $|\cdot|$ denotes the cardinality of a set, i.e., the number of elements in the set.

To mitigate the potential for trivial solutions that achieve full coverage by predicting all available classes, we use the average set size. This metric evaluates the efficiency of our ConE-AFC method by measuring the size of the predicted class sets. A smaller average set size, while maintaining the same coverage, suggests a prediction set that is more accurate and efficient. The term is defined as follows:

$$\text{Average Set Size} = 1/N \sum_{n=1}^N |\hat{Y}_n|. \quad (8)$$

Our experimental design involves measuring the performance of all examined AFC methods across each available segment. This means that an individual value is computed for

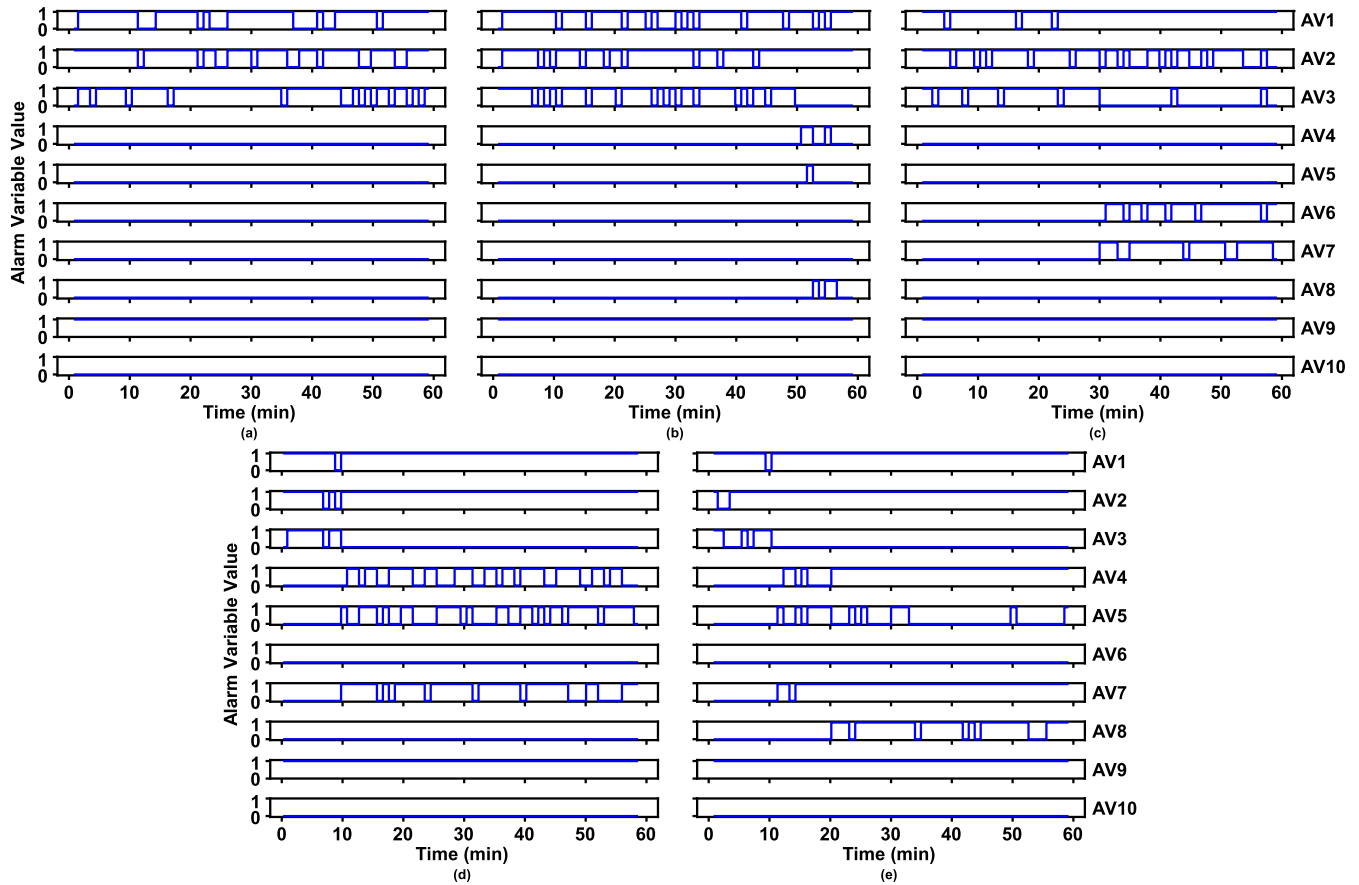


FIGURE 7. Typical alarm series representations of alarm variables AV1 to AV10 for the alarm subsequence classes C1 to C5 in the synthetic alarm evaluation dataset. The solid blue lines represent alarm variable time trends. A higher level represents an active alarm. (a) Alarm subsequence class C1. (b) Alarm subsequence class C2. (c) Alarm subsequence class C3. (d) Alarm subsequence class C4. (e) Alarm subsequence class C5.

each expanding window, where the set of all windows T can be described as $T = \{T(t_i) \mid t_i = i \times \omega, i = 1, 2, \dots, \lfloor L/\omega \rfloor\}$, with L denoting the length of the subsequences in the evaluation dataset.

C. EXPERIMENTAL SETUP

Five existing AFC methods from literature described in Section II are examined in our evaluation. The first method, as described in reference [5], utilizes characteristic alarm coactivations and SVMs for AFC. Henceforth, this method is referred to as ACM-SVM. The second method, CASIM, which is based on convolutional kernels, was previously proposed in [48]. Furthermore, we investigate the efficacy of the dissimilarity index-based method, as described in reference [33], henceforth referred to as WDI-1NN. The fourth method utilizes the AFC method described in [38], which employs EAC vector representations and a 1NN classifier that relies on the distances incorporated in a constructed k-d tree. Henceforth, this AFC method is referred to as EAC-1NN. The fifth method, as outlined in [18], uses MBW, TF-IDF, and a weighting strategy to incorporate key characteristics from alarm sequences into a feature vector. Subsequently, it utilizes

a one-vs-rest approach to train binary logistic regression classifiers. Henceforth, this method is referred to as MBW-LR.

These five AFC methods are selected due to their alignment with the inputs of our ConE-AFC method, namely posterior class probabilities (ACM-SVM, CASIM, and MBW-LR) and distance measures (WDI-1NN and EAC-1NN). Additionally, these methods offer distinct perspectives on alarm dynamics, specifically alarm sets (WDI-1NN), sequences (EAC-1NN and MBW-LR), and series (ACM-SVM and CASIM). The AFC performance of these five methods is compared, both with and without the application of our proposed ConE-AFC method, to the AFC models.

In addition, our ConE-AFC method is evaluated against an alternative version of it that substitutes the ETSC-based incremental calibration with a calibration method that considers the complete length of the historical alarm subsequences. This alternative approach, henceforth referred to as “conformal AFC” (Con-AFC) method, does not fully account for temporary class ambiguity, but rather focuses on class ambiguity that prevails over the entire alarm dynamics. This evaluation approach permits a comprehensive evaluation of the ConE-AFC method’s effectiveness, as well as its ETSC and CP integration.

The purpose of this evaluation is not to perform extensive hyperparameter tuning, but rather to demonstrate the utilization of CP and ETSC in online AFC and the potential of our proposed ConE-AFC method to enhance existing pre-trained AFC models. Therefore, we employ default parameters derived from existing literature, as appropriate, and refrain from evaluating the performance of the novelty detection stages in the AFC models.

For ACM-SVM, the parameters of Platt's probabilistic output to the SVMs are tuned according to [100]. The parameters that undergo optimization through cross-validation are adjusted for each individual training-test set, rendering them dynamic and not explicitly defined in this context, as described in [5]. According to [5], the ACM-SVM model is trained using the complete length of historical alarm subsequences. For online AFC, the evolving alarm data is used as input in an incremental manner as it becomes available.

For WDI-1NN, we implement the method according to the detailed description given in [33]. A template alarm set vector is generated for each historical alarm subsequence class, containing only those alarm variables active in at least 50% of the subsequences within the class. However, since [33] does not provide an online implementation of WDI-1NN, we employ an expanding window strategy, as depicted in Fig. 8, which closely resembles the training approach utilized in [48] for CASIM. For each window $T(t_i)$ an individual AFC model is trained consisting of a transformation stage $ts(t_i)$ and a classification stage $cs(t_i)$. Here, an interval length ω of 1 minute is used, meaning that a new template is generated for each available time step with the initial window, denoted as $T(t_1)$, having a length 10 minutes. The latter option was chosen due to the conventional practice of detecting the beginning of an alarm flood by assessing the frequency of alarm activations within a 10-minute timeframe [16].

For CASIM, it is necessary to establish the appropriate number of features for the alarm series transformation. The evaluation findings presented in [48] indicate that a minimum of 672 features is viable, even when applied to industrial processes of greater complexity than the synthetic dataset utilized in this study. Moreover, tests in [48] also showed that using a single classifier instead of a classifier ensemble is a suitable setting for achieving an acceptable balance between efficiency and high accuracy. Following [48], we utilize an expanding window approach to train the online implementation of CASIM. In order to facilitate comparison, we employ the identical experimental configuration as that used for WDI-1NN in Fig. 8.

Following [38], EAC-1NN's process-specific attenuation coefficient λ is set to 0.0667/min. For MBW-LR, the individual class thresholds are computed for each test based on the 95% confidence interval of the respective Gaussian distribution of training probability estimates [18]. The parameter θ for the binary logistic regression classifiers is obtained according to [101] during training using a one-versus-rest strategy. Furthermore, both EAC-1NN and MBW-LR are

trained using expanding windows that are comparable to those used in WDI-1NN and CASIM.

In our proposed ConE-AFC method, it is necessary to define the parameters α , δ , as well as n_k for each $k \in K$. These parameters will be used to calculate ϵ . In accordance with the recommendations provided in [23] and [94], we set $\delta = 0.1$ and experiment with various settings for n_k , ensuring that we consistently use the same n_k for each $k \in K$ for the purpose of comparison. Three settings for n_k are chosen, namely 22, with an ϵ of 0.1, 102, with an ϵ of 0.05, and 2491, with an ϵ of 0.01 [23]. In addition, we select three settings, namely 0.1, 0.05, and 0.01, for α , which is expected to have the greatest influence on the coverage and average set size [23]. For the expanding windows utilized in the ConE-AFC method, we use the same settings as for WDI-1NN and CASIM. Specifically, we utilize an interval length ω of 1 minute, with the first window having a duration of 10 minutes.

Then, using a stratified 5-fold cross-validation we divide the 18,750 alarm subsequences of the evaluation dataset into five random sets, each containing a training, a calibration, and test split in which the relative frequency of the alarm subsequence classes is maintained between the three splits. To ensure robustness and reliability of our results, we carry out 10 repetitions of the 5-fold cross validation, each time using a unique random seed to account for variability. Consequently, we perform our evaluation on 50 randomly compiled sets from our dataset. In order to ensure comparability, all five examined AFC methods, both with and without our ConE-AFC method being applied, yield identical splits, even if not all of the calibration data is required. As a result, for any of the five sets the same training and test data is used for each AFC model, allowing for a thorough investigation. Fig. 9 illustrates the three splits along with the corresponding count of alarm subsequences within each split, both per class and in total.

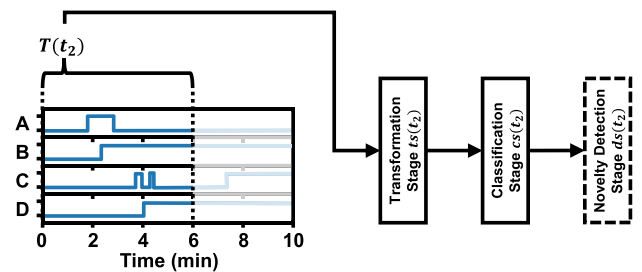


FIGURE 8. Concept of expanding windows for training the WDI-1NN and CASIM online alarm flood classification for exemplary alarm data. With window T , step t , and interval length ω . For each window $T(t_i)$ an AFC model is trained consisting of a transformation stage $ts(t_i)$, a classification stage $cs(t_i)$, and a novelty detection stage $ds(t_i)$. The latter is not examined in this study.

Furthermore, Fig. 10(a) illustrates the expected average accuracy that can be achieved by a conventional AFC model with a single class label output. Fig. 10(b) shows the expected average set size that can be attained through the implementation of our proposed ConE-AFC method on an AFC model capable of capturing all relevant alarm dynamics. When we

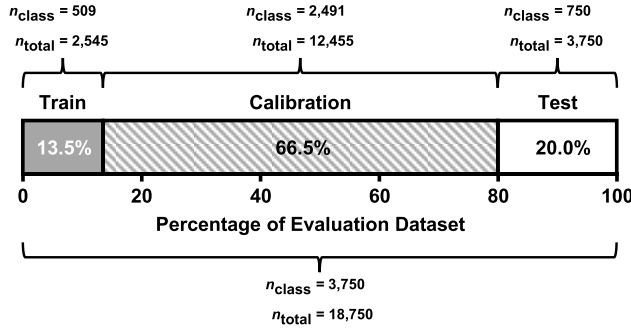


FIGURE 9. The evaluation dataset is divided into three splits: training, calibration, and test. For each split, the percentage of alarm subsequences, as well as the number of subsequences per class n_{class} and the total number of subsequences n_{total} is shown.

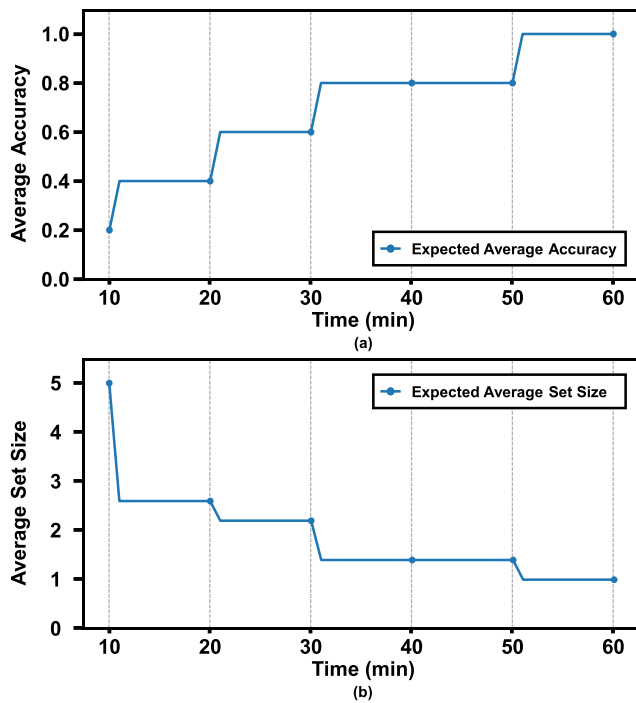


FIGURE 10. Expected performance metrics that can be achieved by the examined AFC methods on the synthetic evaluation dataset. (a) Expected average accuracy for AFC models with a single class label output. (b) Expected average set size for the application of the ConE-AFC method.

calculate the mean value across all time steps, the expected average coverage is 0.710 and the average set size is 1.784. Both expected metrics take into account the experimental setup and the temporary class ambiguities present in our synthetic alarm dataset.

The computational experiments are conducted on a 64-bit Windows PC with an Intel(R) Core(TM) i7-7700HQ 2.80 GHz CPU and 16.0 GB memory. The ConE-AFC method and all AFC methods examined here are implemented in Python (3.9.7) using NumPy [102] (1.22.4), Pandas [103] (1.5.1), sktime (0.13.4) [104], and Scikit-learn [105] (1.1.3) as additional libraries. Our novel evaluation dataset, the method-agnostic implementation of our proposed ConE-AFC

method, as well as all examined AFC methods are made publicly available.¹

D. EVALUATION RESULTS AND DISCUSSION

1) OVERVIEW

The results of our evaluation demonstrate that the WDI-1NN, ACM-SVM, CASIM, EAC-1NN, and MBW-LR methods exhibit different levels of performance when applied to the synthetic evaluation dataset. This is illustrated in Fig. 11, which displays the average accuracy across the test splits of the repeated 5-fold cross validation and each time step. A more detailed numerical analysis of the evaluation results is presented in Table 1, which focuses on coverage, and Table 2, which focuses on set size. These tables provide the average performance metrics across all tests and time steps, offering a clearer comparison of the different parameter settings and the AFC methods to which our proposed ConE-AFC method was applied. Both Table 1 and Table 2 display the mean performance and the standard deviation, calculated over all 10 repeated stratified 5-fold cross validations, resulting in a total of 50 tests.

Additionally, Fig. 12 displays the average distances (WDI-1NN and EAC-1NN) and posterior class probabilities (ACM-SVM, CASIM, and MBW-LR) for each class and time step. The calculation of the latter involves determining the distance or probability, respectively, of the true class of the classified alarm subsequence sample, regardless of the most likely class determined by the respective AFC model. Mathematically, the average posterior probability is defined as:

$$\text{Average Posterior Probability} = \sum_{n=1}^N P(y_n|x_n) / N, \quad (9)$$

where N is the total number of alarm subsequences in the test dataset, y_n is the true class of the n -th alarm subsequence x_n , and $P(y_n|x_n)$ is the posterior probability of the true class given by the classification model. Similarly, the average distance can be calculated as follows:

$$\text{Average Distance} = \sum_{n=1}^N d(y_n, x_n) / N, \quad (10)$$

where $d(y_n, x_n)$ is the distance assigned by the classification model to the true class label y_n for the n -th subsequence x_n .

2) BENCHMARK RESULTS

As depicted in Fig. 11(a), WDI-1NN encounters difficulties in achieving accuracy beyond mere guessing, i.e., an accuracy of 0.2. Nevertheless, this approach demonstrates temporary improvements, with accuracy reaching peaks between 0.4 and 0.5 after class divisions at 10, 20, 30, and 50 minutes, only to subsequently decrease in the majority of instances. Upon closer analysis, considering the data presented in Fig. 12(a), it becomes evident that these phases are associated with higher class distances in some classes. This phenomenon can be attributed to the utilization of probabilistic alarm set

¹<https://doi.org/10.24433/CO.5512337.v2>

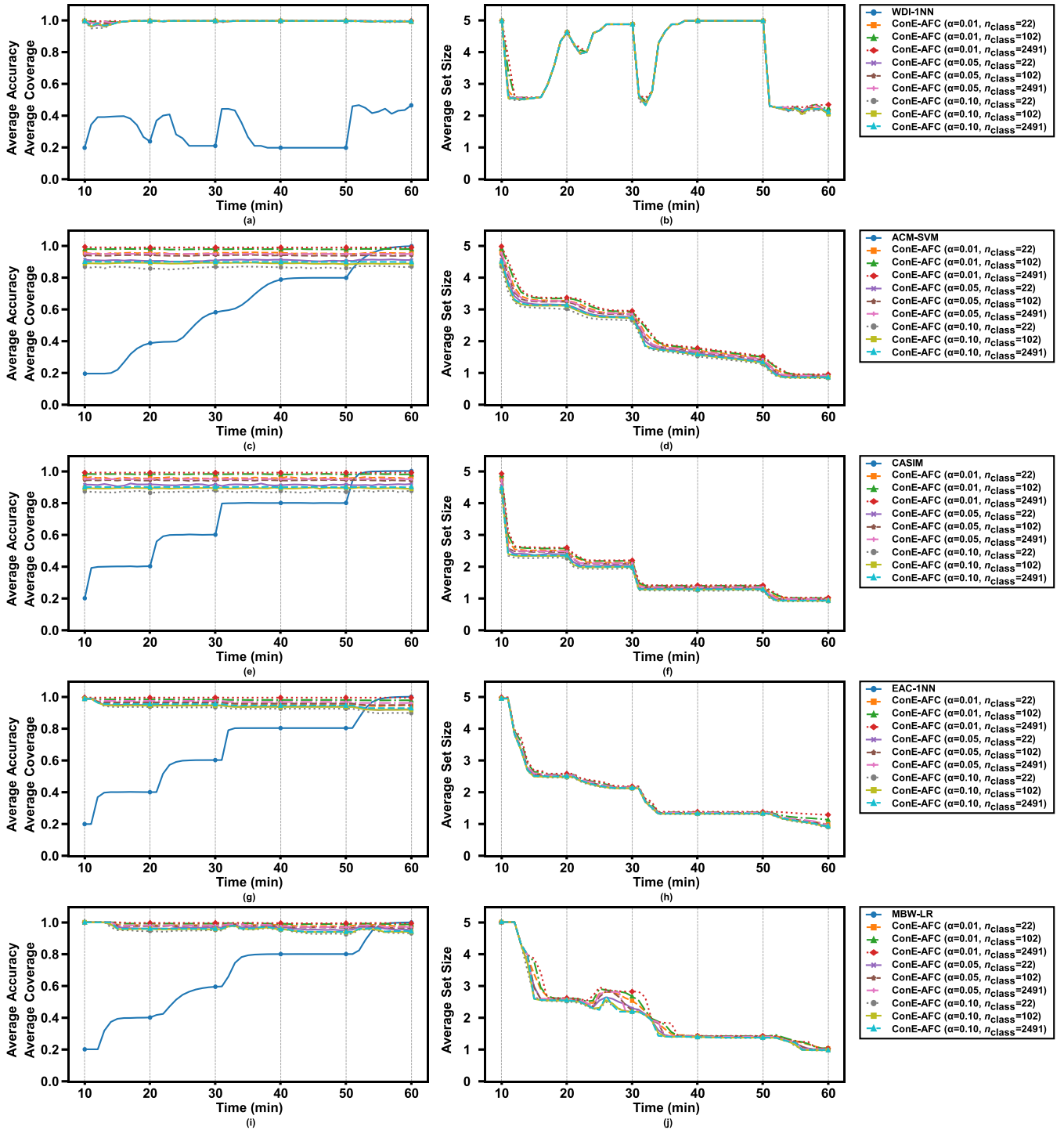


FIGURE 11. Comparative performance analysis of AFC methods over time and across test splits. (a) WDI-1NN's average accuracy and coverage, with and without ConE-AFC application. (b) Average set size resulting from ConE-AFC's application to WDI-1NN. (c) ACM-SVM's average accuracy and coverage, with and without ConE-AFC application. (d) Average set size resulting from ConE-AFC's application to ACM-SVM. (e) CASIM's average accuracy and coverage, with and without ConE-AFC application. (f) Average set size resulting from ConE-AFC's application to CASIM. (g) EAC-1NN's average accuracy and coverage, with and without ConE-AFC application. (h) Average set size resulting from ConE-AFC's application to EAC-1NN. (i) MBW-LR's average accuracy and coverage, with and without ConE-AFC application. (j) Average set size resulting from ConE-AFC's application to MBW-LR.

thresholds, including newly activated alarm variables when they surpass a relative frequency of 50% in the training data. However, in the test split, not all subsequences trigger these

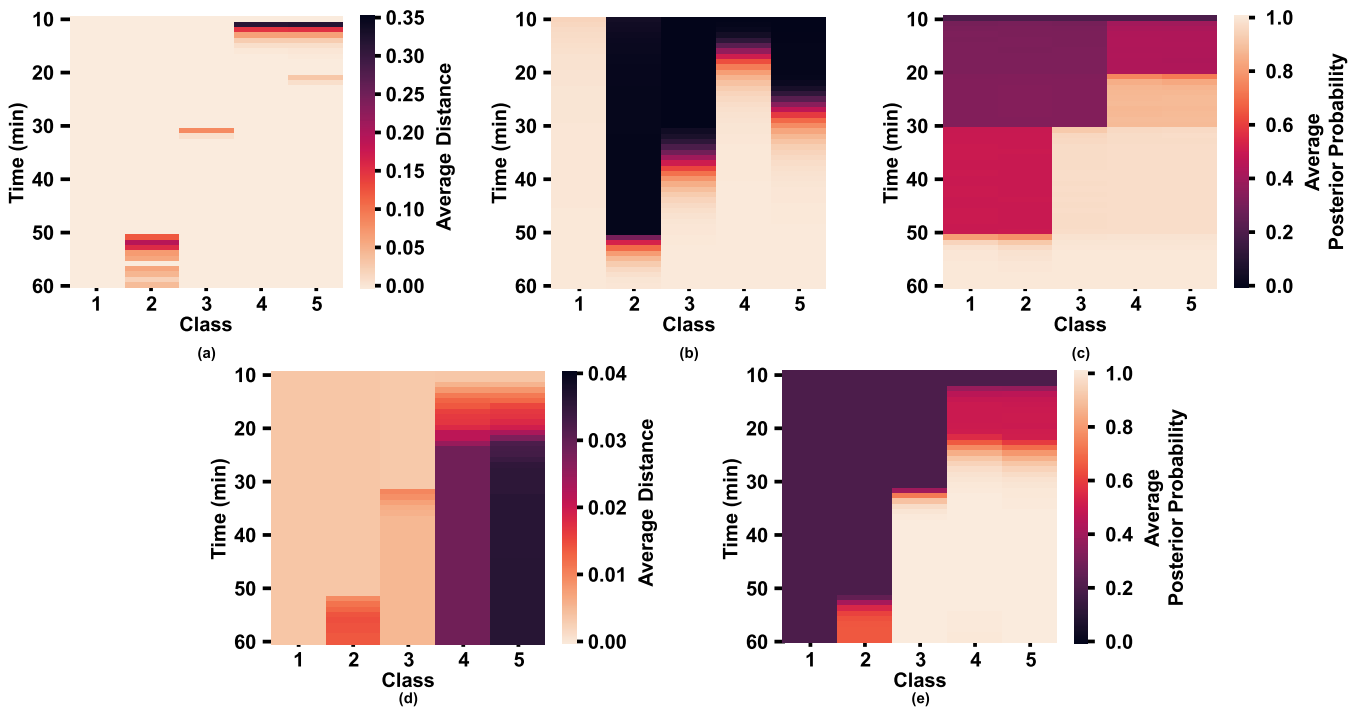
alarms at an early stage, resulting in an increased distance to the correct template. Although this impedes the accurate classification of these classes, it also enables a better

TABLE 1. Average accuracy and coverage over all tests and time steps.

AFC Model	Original	ConE-AFC $\alpha=0.01$ $n_{\text{class}}=22$	ConE-AFC $\alpha=0.01$ $n_{\text{class}}=102$	ConE-AFC $\alpha=0.01$ $n_{\text{class}}=2491$	ConE-AFC $\alpha=0.05$ $n_{\text{class}}=22$	ConE-AFC $\alpha=0.05$ $n_{\text{class}}=102$	ConE-AFC $\alpha=0.05$ $n_{\text{class}}=2491$	ConE-AFC $\alpha=0.10$ $n_{\text{class}}=22$	ConE-AFC $\alpha=0.10$ $n_{\text{class}}=102$	ConE-AFC $\alpha=0.10$ $n_{\text{class}}=2491$
WDI-1NN	0.313±0.019	0.998±0.001	0.999±0.000	1.000±0.000	0.997±0.001	0.998±0.000	0.999±0.000	0.995±0.001	0.997±0.001	0.997±0.000
ACM-SVM	0.629±0.001	0.954±0.008	0.980±0.004	0.990±0.001	0.909±0.012	0.940±0.007	0.950±0.002	0.865±0.015	0.891±0.008	0.900±0.003
CASIM	0.706±0.002	0.957±0.004	0.980±0.002	0.990±0.000	0.914±0.007	0.942±0.003	0.950±0.001	0.871±0.006	0.892±0.003	0.900±0.001
EAC-1NN	0.684±0.001	0.953±0.014	0.977±0.006	0.990±0.002	0.937±0.016	0.956±0.007	0.965±0.003	0.924±0.015	0.935±0.008	0.944±0.004
MBW-LR	0.665±0.001	0.983±0.004	0.992±0.001	0.996±0.000	0.966±0.005	0.976±0.003	0.980±0.001	0.949±0.007	0.957±0.004	0.960±0.002

TABLE 2. Average set size over all tests and time steps.

AFC Model	ConE-AFC $\alpha=0.01$ $n_{\text{class}}=22$	ConE-AFC $\alpha=0.01$ $n_{\text{class}}=102$	ConE-AFC $\alpha=0.01$ $n_{\text{class}}=2491$	ConE-AFC $\alpha=0.05$ $n_{\text{class}}=22$	ConE-AFC $\alpha=0.05$ $n_{\text{class}}=102$	ConE-AFC $\alpha=0.05$ $n_{\text{class}}=2491$	ConE-AFC $\alpha=0.10$ $n_{\text{class}}=22$	ConE-AFC $\alpha=0.10$ $n_{\text{class}}=102$	ConE-AFC $\alpha=0.10$ $n_{\text{class}}=2491$
WDI-1NN	3.850±0.138	3.877±0.141	3.896±0.138	3.837±0.139	3.848±0.140	3.851±0.139	3.826±0.138	3.836±0.138	3.843±0.139
ACM-SVM	2.255±0.045	2.321±0.036	2.356±0.034	2.134±0.044	2.202±0.032	2.224±0.027	2.031±0.047	2.086±0.029	2.102±0.023
CASIM	1.731±0.011	1.778±0.008	1.800±0.004	1.640±0.013	1.691±0.006	1.705±0.002	1.559±0.013	1.599±0.007	1.612±0.002
EAC-1NN	1.906±0.025	1.954±0.022	2.000±0.016	1.879±0.023	1.904±0.015	1.918±0.011	1.859±0.022	1.872±0.011	1.882±0.009
MBW-LR	2.128±0.046	2.168±0.033	2.206±0.010	2.039±0.030	2.067±0.016	2.064±0.006	1.980±0.023	1.993±0.019	2.000±0.009

**FIGURE 12.** Class-specific distance and probability heatmaps over time. Illustrating the average metrics for true class distances and probabilities during alarm subsequence classification, independent of the predicted class by the AFC model. (a) WDI-1NN's distance heatmap. (b) ACM-SVM's posterior class probability heatmap. (c) CASIM's posterior class probability heatmap. (d) EAC-1NN's distance heatmap. (e) MBW-LR's posterior class probability heatmap.

differentiation between other classes. However, WDI-1NN's poor overall performance can be explained by the limited number of alarm variables available. At certain points, most of these variables are activated in all classes, resulting in

distances close to zero. In other words, the alarm sets are identical across most classes.

Table 1 highlights the poor performance of WDI-1NN, revealing that when applied without our proposed ConE-AFC

method, it achieves a low average accuracy of 0.313 over all 50 tests and time steps. This accuracy comes with a relatively high standard deviation of 0.019, which is significantly greater than the standard deviations observed for the other examined AFC methods, which range between 0.001 and 0.002. This variability further underscores the inconsistency of WDI-1NN in effectively classifying alarm subsequences that show significant ambiguities.

Fig. 11(c) illustrates the limited performance of ACM-SVM in dealing with temporary class ambiguities and its gradual adaptation to new class distinctions. This is indicated by the smooth and significant increases in accuracy that occur after each class separation event, e.g., at 10, 20, 30, and 50 minutes. These adaptations mirror the expected outcomes, shown in Fig. 10(a), based on the synthetic dataset's designed temporary class ambiguities. Fig. 12(b) provides further insight into this by showcasing the evolution of average posterior class probabilities across different time windows and classes, indicating a slow but steady increase in probability for the newly distinguishable class. However, the ACM-SVM exhibits biases towards specific classes, even though the dataset is balanced. This phenomenon, particularly pronounced prior to class transitions when class probabilities approach zero for the majority of classes, may mislead operators by suggesting unjustified certainty in class predictions amid substantial class ambiguity.

The improved performance of ACM-SVM compared to WDI-1NN is also reflected in Table 1, where it achieves an average accuracy of 0.629 with a standard deviation of 0.001. However, this still represents the second-lowest performance among the five examined benchmark AFC methods. In comparison to CASIM, EAC-1NN, and MBW-LR, which all utilize expanding window approaches that specifically learn and capture local alarm dynamics in a more targeted manner, ACM-SVM's approach of training only once on the full dynamics limits its ability to focus on more local distinctions. This limitation emphasizes the reason why ACM-SVM is less effective than other methods in adapting to the evolving alarm subsequences.

CASIM, which uses expanding windows for training AFC classifiers, adapts to new class patterns more quickly compared to ACM-SVM. This is evident from the abrupt improvements in accuracy after the separation of classes (Fig. 11(e)), which match the expected accuracies, as shown in Fig. 10(a), closely. The relatively strong performance of CASIM is further evidenced in Table 1, where it achieves an average accuracy of 0.706 with a standard deviation of 0.002. This represents the best performance among all the benchmark AFC methods examined and approaches the expected achievable average accuracy of 0.710 depicted in Fig. 10(a).

Fig. 12(c) illustrates that the average posterior class probabilities for CASIM show a rapid rise to significantly higher probabilities post-separation, surpassing ACM-SVM in terms of response time. Furthermore, we observe that in the periods of present temporary class ambiguity, the probability is divided among the classes that are ambigu-

ous. This may already contain relevant information for the operator demonstrating that the AFC model lacks the capability to definitively ascertain a single class. Nevertheless, CASIM remains susceptible to these uncertainties, and in such instances, it delivers unreliable predictions.

EAC-1NN demonstrates a dynamic performance similar to that of CASIM, as seen in Fig. 11(g). However, it tends to take slightly longer to adjust to the new available dynamics and learn meaningful characteristics, as evidenced by the slower increase in accuracy following key time points after class division at 10, 20, 30, and 50 minutes. This delay may be attributed to EAC-1NN's emphasis on early alarms and the reliance on the 1NN classifier, which might require more time for enough alarm variables to exhibit altered behavior, allowing these changes to be adequately reflected in the distance calculations for the true class.

Despite the performance delay, EAC-1NN performs significantly better than WDI-1NN, the other examined 1NN-based classifier, by learning and distinguishing more viable alarm characteristics, even when dealing with relatively similar classes. This is further illustrated in Fig. 12(d), where the distance to the true class starts close to zero but increases over time as more alarm variables activate at the class junction points. Interestingly, as the distance to the true class grows, an in-depth analysis of the underlying class distances reveals that the distances to other classes increase even more. Nevertheless, the dynamic range of distances is relatively small, ranging between 0.00 and 0.04, yet it proves sufficient for distinguishing the classes as they become more discernible over time.

Table 1 also reflects EAC-1NN's relatively strong performance, where it achieves the second-highest average accuracy of 0.684 across all 50 tests and time steps, with a standard deviation of 0.001. This highlights EAC-1NN's capability to effectively learn and adapt to evolving alarm dynamics. However, EAC-1NN is still subject to temporary class ambiguities, and its performance deteriorates during these phases, leading to less reliable classification outcomes.

MBW-LR exhibits a performance that falls between ACM-SVM and EAC-1NN, as shown in Fig. 11(i). While it manages to improve its classification accuracy when additional characteristics of the temporarily ambiguous classes become available, it does so at a slower pace compared to EAC-1NN and CASIM. Fig. 12(e) further illustrates that the average posterior class probabilities for MBW-LR increase gradually, but not as rapidly or effectively as CASIM. Unlike CASIM, MBW-LR struggles to allocate some probability to the ambiguous classes, which can be critical in handling temporary class ambiguities. For example, even though class C2 becomes distinguishable from class C1 after 50 minutes, the average probability assigned by MBW-LR remains only around 65% for C2 and 20% for C1, indicating its poorer performance in these challenging scenarios. Table 1 supports this observation, showing that MBW-LR achieved an average accuracy of 0.665 and a standard deviation of 0.001 across all tests and time steps, placing it in the middle of the examined

AFC methods. Nevertheless, it exhibits subpar performance when faced with temporary class ambiguity.

3) RESULTS OF THE PROPOSED METHOD

An examination of our proposed ConE-AFC method's performance, when applied to WDI-1NN, ACM-SVM, CASIM, EAC-1NN, and MBW-LR, reveals notable improvements in classification reliability, as evidenced by Figs. 11(a), (c), (e), (g), and (i), respectively. Additionally, our method maintains reasonable average set sizes in cases where the pre-trained AFC model provides sufficient discriminatory insights, as depicted in Figs. 11(b), (d), (f), (h), and (j), respectively. In the subsequent paragraphs, we thoroughly examine the performance of our proposed ConE-AFC method, as illustrated in Fig. 11 and explained in numerical detail in Tables 1 and 2.

For WDI-1NN, Fig. 11(a) indicates that ConE-AFC maintains average coverage over all tests near 1.000 across all parameter settings throughout the observed time span. This high coverage ensures that the true class is almost always included in the set of potential classes predicted, thereby significantly enhancing reliability. This phenomenon is also illustrated in Table 1, where the initial average accuracy across all tests and time steps of 0.313 ± 0.019 improves significantly to achieve almost perfect coverage in all settings, ranging from 0.995 to 1.000.

However, the improvement in coverage when applying ConE-AFC to WDI-1NN is accompanied by substantial average set sizes, particularly in the 25- to 50-minute range, where the set sizes can encompass all five classes, as shown in Fig. 11(b). In addition, Table 2 displays a relatively high average set size over all time steps ranging from 3.826 to 3.896 for WDI-1NN using ConE-AFC, which is nearly twice as large as that of the majority of the other analyzed AFC methods. This suggests that the method frequently predicts almost all classes, which is attributed to the limited discriminative power of WDI-1NN. This observation is further supported by the low overall class distances depicted in Fig. 12(a), which underscores the challenge of predicting non-trivial class sets with WDI-1NN using ConE-AFC, as the method's alarm set templates fail to effectively differentiate between classes.

When applying ConE-AFC to ACM-SVM, a significant improvement in coverage is observed in Fig. 11(c) where the coverage remains significantly above the baseline accuracy up to the 50-minute mark. Moreover, the average coverage over all time steps and tests increases from an original accuracy of 0.629 ± 0.001 to a range of 0.865–0.990 across various parameter settings (Table 1). This enhancement demonstrates ConE-AFC's effectiveness in addressing the temporary class ambiguities that challenge ACM-SVM. However, if ACM-SVM is used without ConE-AFC it shows an improved performance towards the end of the observed alarm subsequences in Fig. 11(c). Other AFC methods that do not implement ConE-AFC display a similar behavior. This phenomenon might be due to the

resolution of temporary class ambiguities, whereas ConE-AFC's statistical thresholding may occasionally omit the true class for subsets of the test data, particularly when posterior probabilities are marginally below the threshold.

As shown in Table 2, when using ConE-AFC on ACM-SVM to improve coverage, there is a trade-off in the form of an increase in the average set size over all time steps and tests. The latter ranges from 2.031 to 2.356. Although this set size increase is less pronounced than in WDI-1NN, it is still higher than the expected average set size over all time steps of 1.784. This indicates that the model's ability to identify a single class with high confidence during periods of uncertainty is still limited. The slightly larger set sizes with lower α values indicate that ConE-AFC accommodates the uncertainty by suggesting multiple potential classes, thus improving overall reliability at the cost of precision.

CASIM demonstrates the best trade-off performance among the tested models when combined with ConE-AFC, as indicated by both high coverage and relatively small average set sizes. The coverage for CASIM increases from an original average accuracy over all tests and time steps of 0.706 ± 0.002 to a range of 0.871–0.990, depending on the parameter settings (Table 1). This reflects CASIM's strong adaptability to evolving alarm dynamics, further enhanced by ConE-AFC's ability to manage early-stage ambiguities. Notably, CASIM maintains a lower average set size over all tests and time steps compared to the other models, ranging from 1.559 to 1.800 (Table 2), which suggests that CASIM, with ConE-AFC, can effectively narrow down the potential classes with greater precision, thus providing operators with more actionable and focused predictions.

Moreover, the coverage and average set size for CASIM, depicted in Figs. 11(e) and 11(f), respectively, follow the expected patterns of temporary class ambiguity, shown in Fig. 10. The set sizes decrease over time as classes become more distinguishable. Notably, for time steps between 11 and 50 minutes, CASIM shows average set sizes between 0.2 and 1.4 lower than ACM-SVM over all examined parameter settings. Furthermore, during time steps 11 to 13 for ACM-SVM and EAC-1NN, and 11 to 15 for MBW-LR, CASIM demonstrates average set sizes that are more than 1.0 lower. This suggests that the operator receives a much greater amount of useful assistance, especially during the early stages when there is a lot of temporary class ambiguity, by combining CASIM and our proposed ConE-AFC. This phenomenon may be attributed to CASIM's enhanced ability to detect local discriminatory alarm patterns by utilizing a combination of convolutional operations and an expanding window strategy.

When implementing the proposed ConE-AFC on EAC-1NN and MBW-LR, there is a substantial increase in the average coverage across all test and time steps. The average coverage rises from 0.684 ± 0.001 to 0.924–0.990 for EAC-1NN, and from 0.665 ± 0.001 to 0.949–0.996 for MBW-LR, as shown in Table 1. Additionally, both EAC-1NN and MBW-LR exhibit an average set size across all tests and

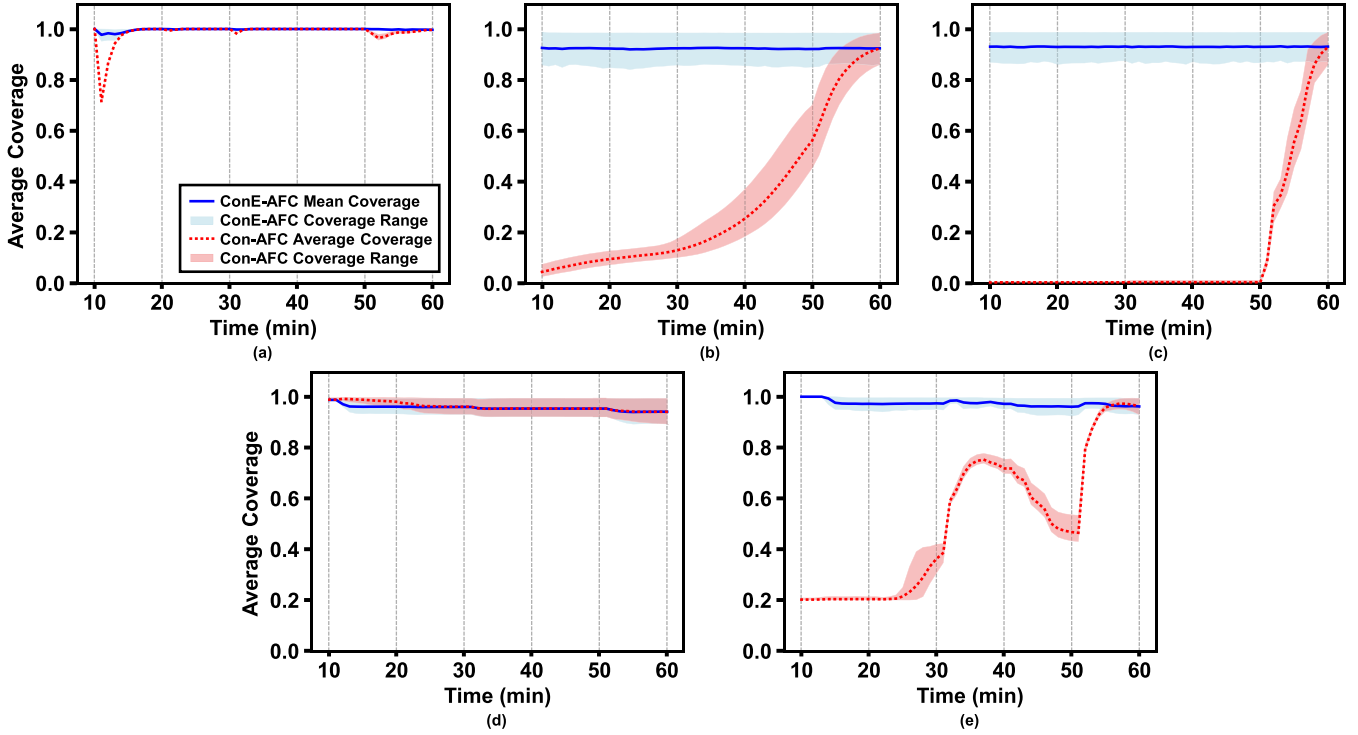


FIGURE 13. Performance comparison of ConE-AFC and Con-AFC methods when applied to pre-trained AFC models over time. Displayed are the average coverages across nine parameter settings (a) Mean coverage and range for ConE-AFC and Con-AFC applied to WDI-1NN. (b) Mean coverage and range for ConE-AFC and Con-AFC applied to ACM-SVM. (c) Mean coverage and range for ConE-AFC and Con-AFC applied to CASIM. (d) Mean coverage and range for ConE-AFC and Con-AFC applied to EAC-1NN. (e) Mean coverage and range for ConE-AFC and Con-AFC applied to MBW-LR.

time steps ranging from 1.859 to 2.000 and 1.980 to 2.206, respectively (Table 2). These values are relatively close to the expected average set size of 1.784.

Moreover, EAC-1NN and MBW-LR exhibit greater resilience to variations in parameter configurations when compared to CASIM and ACM-SVM, particularly during the initial 50 time steps as depicted in Fig. 11. This is evident from the smaller variations in performance across the tested parameter settings. It is worth noting that both EAC-1NN and MBW-LR have a coverage of 1.000 during the initial minutes of classification in Figs. 11(g) and 11(l), respectively. However, this is only possible by including all five classes in the predicted sets, as shown by the average set size in Figs. 11(h) and 11(m), respectively. This indicates that there is significant ambiguity for both AFC methods, which are unable to differentiate between any classes, at the start of the evolving alarm subsequences.

Fig. 13 illustrates a comparative assessment of the proposed ConE-AFC method in relation to its variant, Con-AFC, which does not employ expanding windows during the calibration stage. The comparative analysis is based on the average and range of coverages across all nine tested parameter settings and across all five examined AFC methods.

The INN classifiers WDI-1NN and EAC-1NN, shown in Figs. 13(a) and 13(d) respectively, exhibit comparable behavior. The performance of both ConE-AFC and Con-AFC is similar, except for brief periods when either variant

demonstrates an advantage. The impact of evolving dynamics on INN calculations is less significant compared to posterior class probabilities. This is because the vectors used to compute the distances do not undergo significant changes over time, as shown by the relatively limited dynamic range of distances depicted in Figs. 12(a) and (d).

The three AFC methods analyzed, namely ACM-SVM (Fig. 13(b)), CASIM (Fig. 13(c)), and MBW-LR (Fig. 13(e)), which use posterior class probabilities, exhibit notable distinctions between the proposed ConE-AFC and its variant Con-AFC. In terms of coverage, the ConE-AFC method consistently demonstrates superior performance compared to Con-AFC, which exhibits a rather subpar performance until the end. The reason for this is that Con-AFC is calibrated exclusively using the full-length alarm dynamics of alarm subsequences in the calibration dataset. The resulting posterior class probabilities, however, can exhibit substantial variations between the full length and any portion of it, as demonstrated by the broad range of average probabilities shown in Figs. 12(b), (c), and (e). As a result, Con-AFC's predictive accuracy is limited during the early stages of the tested alarm subsequences. On the other hand, ConE-AFC's improvement can be primarily attributed to the implementation of expanding windows during the calibration stage. This enables ConE-AFC to effectively adapt to temporary class ambiguities that may arise during the initial stages of an alarm subsequence.

4) DISCUSSION SUMMARY

The evaluation results indicate that none of the existing and examined AFC methods can independently manage temporary class ambiguities effectively. When two or more classes cannot be distinguished, these methods often default to predicting a single dominant class based on biases from the training process or resort to random guessing influenced by their decision boundaries. Additionally, several methods, such as ACM-SVM, EAC-1NN, and MBW-LR, exhibit delays in adapting to newly emerging characteristics that distinguish classes, leading to unreliable predictions that may not provide timely support for human operators.

By effectively managing uncertainty and ambiguity throughout the duration of evolving alarm subsequences, our proposed ConE-AFC method significantly improves the performance of the analyzed AFC models. Our method ensures higher coverage, consistently including the true class in the predicted sets, which improves reliability. While this increase in coverage may lead to larger set sizes, indicating a trade-off between reliability and precision, models like CASIM demonstrate a more favorable balance with robust adaptability and smaller set sizes. In contrast, models such as WDI-1NN, ACM-SVM, EAC-1NN, and MBW-LR exhibit comparable coverages, but with the drawback of larger set sizes and slower adaptability to class ambiguities as they evolve dynamically. Although ConE-AFC offers a dynamic and method-agnostic approach to AFC that can ensure high coverage, its balance between coverage and set size is heavily influenced by the quality of the underlying AFC model to which ConE-AFC is implemented, which adheres to the “Garbage-In, Garbage-Out” principle in computer science [106].

Our analysis also reveals an inversely proportional relationship between α values and average set sizes in ConE-AFC; smaller α values result in larger average set sizes, reflecting greater uncertainty in classification. For n_{class} , our evaluation results in Table 2 demonstrate a proportional relationship with the average set size.

Furthermore, ACM-SVM, CASIM, EAC-1NN, and MBW-LR often produce average set sizes below 1.000 in the final 10-minute window, as shown in Fig. 11, suggesting that ConE-AFC’s thresholds might yield empty prediction sets when no posterior class probabilities exceed the threshold. To mitigate this, practitioners could implement a fallback strategy that defaults to predicting the most likely class when an empty set is encountered, ensuring that operators receive consistent guidance even in uncertain scenarios.

V. CONCLUSION AND OUTLOOK

Our comprehensive evaluation in Section IV reveals that existing AFC methods, such as WDI-1NN, ACM-SVM, CASIM, EAC-1NN, and MBW-LR exhibit substantial limitations under conditions of temporary class ambiguity, impairing their ability to reliably discriminate between classes as alarm floods and subsequences evolve. Moreover, because

they rely on narrow distance measures (WDI-1NN and EAC-1NN) or biased posterior class probabilities (ACM-SVM, CASIM, and MBW-LR) to make single-label predictions, none of the examined AFC methods from the literature provides the capability to offer direct insights regarding the dynamic likelihood of different but ambiguous classes.

Contrastingly, our novel ConE-AFC method, which introduces the first attempt to synergistically integrate CP and ETSC into AFC, demonstrates significant advantages. It consistently provides operators with a dynamic and updated set of likely alarm subsequence classes, thereby offering a finely tunable trade-off between coverage and the size of prediction sets to meet specific industrial requirements. By delivering accurate, multi-label classifications with clear confidence indicators, operators are better equipped to prioritize and respond to alarms, potentially mitigating the risks associated with alarm floods. We argue that this explainability could enhance trust and hence promote the acceptance of AI and ML solutions in critical industrial environments, where the importance of operational safety and efficiency is significant [23], [29], [30], [31], [32].

A notable limitation of our approach is the necessary amount of data for the calibration set, which directly correlates with the coverage of class predictions. However, the size of the calibration set in our study had a limited effect on performance, as low numbers of subsequences per class in the calibration set already provided adequate coverage. Nevertheless, in real-world scenarios, these quantities of historical abnormal situations may not always be readily available. Therefore, future studies should attempt to artificially create a calibration dataset that accurately reflects realistic alarm dynamics. This could be achieved by employing techniques such as time series data augmentation, digital twins, and process simulators.

Future research could further explore the domain of explainable artificial intelligence techniques in order to explain the underlying reasoning behind predicted alarm subsequence class sets, hence improving operators’ understanding of temporary class ambiguity and model uncertainty [32]. One potential approach to accomplish this is by adapting our previously proposed explainable AI approach, which was specifically designed for single-label AFC in [20], to suit the multi-label predictions generated by our ConE-AFC method. Additionally, future research could explore whether the trigger functions from ETSC, which determine when sufficient data is observed to differentiate classes [81], can also be used to extract further insights about the specific moments when temporary class ambiguities are resolved. Such insights could help operators understand which changes in the underlying alarm dynamics have the most significant influence on class differentiation, providing more effective and trustworthy decision support.

Furthermore, this study did not investigate the novelty detection stage in AFC methods, which detects new classes that have not been encountered before. This stage is crucial, as ML-based classifiers may incorrectly claim a high level of

confidence in class predictions even if they are beyond the range of their training. Further study is necessary to investigate the integration of OOD detection and our proposed ConE-AFC method within a unified AFC framework.

Finally, in this initial investigation, we have carried out the first stage of design science research [96], which involves conducting a controlled experiment utilizing synthetic data that we possess complete ground-truth knowledge of. This methodology was essential in order to have a comprehensive understanding of temporary class ambiguities for a thorough evaluation. The availability of this type of data and knowledge is limited whether it comes to real-world datasets or datasets generated by process simulators [5], [19]. Nevertheless, previous studies conducted by [17] and [19] have already demonstrated the presence of temporary class ambiguity in more practical industrial contexts. Further investigation in future studies should involve the examination of more complex evaluation datasets.

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CODE AND DATA AVAILABILITY

The study's conclusions are substantiated by publicly accessible and archived code and data. Our proposed ConE-AFC method, along with all five examined AFC methods and our evaluation dataset, have been publicly archived. Since no code for the examined AFC methods was available, we implemented these methods ourselves and have made them publicly accessible. The archive encompasses several components, such as the Python implementation, virtual environment setup, evaluation data, ground-truth labeling, and reproducible results for selected tests of the conducted repeated stratified 5-fold cross-validation. All these can be accessed via the following DOI hyperlink: <https://doi.org/10.24433/CO.5512337.v2>.

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